In [ ]:
```
GSE118389_counts_rsem.txt
```

In [5]:
```
!python -m pip uninstall -y zarr
```

```
Found existing installation: zarr 3.1.6
Uninstalling zarr-3.1.6:
  Successfully uninstalled zarr-3.1.6
```

In [6]:
```
!python -m pip install "zarr<3"
```

```
Collecting zarr<3
  Downloading zarr-2.18.7-py3-none-any.whl.metadata (5.8 kB)
Collecting asciitree (from zarr<3)
  Downloading asciitree-0.3.3.tar.gz (4.0 kB)
  Installing build dependencies: started
  Installing build dependencies: finished with status 'done'
  Getting requirements to build wheel: started
  Getting requirements to build wheel: finished with status 'done'
  Preparing metadata (pyproject.toml): started
  Preparing metadata (pyproject.toml): finished with status 'done'
Requirement already satisfied: numpy>=1.24 in .\anaconda3\Lib\site-packages (from zar
r<3) (2.4.4)
Collecting fasteners (from zarr<3)
  Downloading fasteners-0.20-py3-none-any.whl.metadata (4.8 kB)
Collecting numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0 (from zarr<3)
  Downloading numcodecs-0.15.1-cp311-cp311-win_amd64.whl.metadata (3.0 kB)
Collecting deprecated (from numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0->zarr<3)
  Downloading deprecated-1.3.1-py2.py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: wrapt<3,>=1.10 in .\anaconda3\Lib\site-packages (from
deprecated->numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0->zarr<3) (1.14.1)
Downloading zarr-2.18.7-py3-none-any.whl (211 kB)
Downloading numcodecs-0.15.1-cp311-cp311-win_amd64.whl (840 kB)
   ---------------------------------------- 0.0/840.0 kB ? eta -:--:--
   ---------------------------------------- 0.0/840.0 kB ? eta -:--:--
   ------------ --------------------------- 262.1/840.0 kB ? eta -:--:--
   ------------------------------------- -- 786.4/840.0 kB 1.3 MB/s eta 0:00:01
   ---------------------------------------- 840.0/840.0 kB 1.4 MB/s  0:00:00
Downloading deprecated-1.3.1-py2.py3-none-any.whl (11 kB)
Downloading fasteners-0.20-py3-none-any.whl (18 kB)
Building wheels for collected packages: asciitree
  Building wheel for asciitree (pyproject.toml): started
  Building wheel for asciitree (pyproject.toml): finished with status 'done'
  Created wheel for asciitree: filename=asciitree-0.3.3-py3-none-any.whl size=5112 sh
a256=0d4e610aee07471e0eb64563af24698b25c0d78542f1b6a5ab0e7d8a543c3532
  Stored in directory: c:\users\dgous\appdata\local\pip\cache\wheels\71\c1\da\23077eb
3b87d24d6f3852ed1ed1a1ac2d3c885ad6ebd2b4a07
Successfully built asciitree
Installing collected packages: asciitree, fasteners, deprecated, numcodecs, zarr

   -------- ------------------------------- 1/5 [fasteners]
  Attempting uninstall: numcodecs
   -------- ------------------------------- 1/5 [fasteners]
    Found existing installation: numcodecs 0.16.5
   -------- ------------------------------- 1/5 [fasteners]
   ------------------------ --------------- 3/5 [numcodecs]
    Uninstalling numcodecs-0.16.5:
   ------------------------ --------------- 3/5 [numcodecs]
      Successfully uninstalled numcodecs-0.16.5
   ------------------------ --------------- 3/5 [numcodecs]
   ------------------------ --------------- 3/5 [numcodecs]
   ------------------------ --------------- 3/5 [numcodecs]
   ------------------------ --------------- 3/5 [numcodecs]
   ------------------------ --------------- 3/5 [numcodecs]
   ------------------------ --------------- 3/5 [numcodecs]
   --------------------------------- ------- 4/5 [zarr]
   --------------------------------- ------- 4/5 [zarr]
   --------------------------------- ------- 4/5 [zarr]
   --------------------------------- ------- 4/5 [zarr]
   --------------------------------- ------- 4/5 [zarr]
   ---------------------------------------- 5/5 [zarr]
```

```
Successfully installed asciitree-0.3.3 deprecated-1.3.1 fasteners-0.20 numcodecs-0.1
5.1 zarr-2.18.7
```

In [8]:
```
!python -m pip uninstall -y scanpy anndata h5py numpy zarr
```

```
Found existing installation: scanpy 1.11.5
Uninstalling scanpy-1.11.5:
  Successfully uninstalled scanpy-1.11.5
Found existing installation: anndata 0.12.10
Uninstalling anndata-0.12.10:
  Successfully uninstalled anndata-0.12.10
Found existing installation: h5py 3.16.0
Uninstalling h5py-3.16.0:
  Successfully uninstalled h5py-3.16.0
Found existing installation: numpy 2.4.4
Uninstalling numpy-2.4.4:
  Successfully uninstalled numpy-2.4.4
Found existing installation: zarr 2.18.7
Uninstalling zarr-2.18.7:
  Successfully uninstalled zarr-2.18.7
```

In [9]:
```
!python -m pip install --no-cache-dir "numpy==1.26.4" "h5py==3.11.0" "zarr<3" "scanpy[
```

```
Collecting numpy==1.26.4
  Downloading numpy-1.26.4-cp311-cp311-win_amd64.whl.metadata (61 kB)
Collecting h5py==3.11.0
  Downloading h5py-3.11.0-cp311-cp311-win_amd64.whl.metadata (2.5 kB)
Collecting zarr<3
  Downloading zarr-2.18.7-py3-none-any.whl.metadata (5.8 kB)
Collecting scanpy[leiden]
  Downloading scanpy-1.11.5-py3-none-any.whl.metadata (9.3 kB)
Requirement already satisfied: asciitree in .\anaconda3\Lib\site-packages (from zarr<
3) (0.3.3)
Requirement already satisfied: fasteners in .\anaconda3\Lib\site-packages (from zarr<
3) (0.20)
Requirement already satisfied: numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0 in .\anacond
a3\Lib\site-packages (from zarr<3) (0.15.1)
Requirement already satisfied: deprecated in .\anaconda3\Lib\site-packages (from numc
odecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0->zarr<3) (1.3.1)
Collecting anndata>=0.8 (from scanpy[leiden])
  Downloading anndata-0.12.10-py3-none-any.whl.metadata (9.9 kB)
Requirement already satisfied: joblib in .\anaconda3\Lib\site-packages (from scanpy[l
eiden]) (1.5.3)
Requirement already satisfied: legacy-api-wrap>=1.4.1 in .\anaconda3\Lib\site-package
s (from scanpy[leiden]) (1.5)
Requirement already satisfied: matplotlib>=3.7.5 in .\anaconda3\Lib\site-packages (fr
om scanpy[leiden]) (3.10.8)
Requirement already satisfied: natsort in .\anaconda3\Lib\site-packages (from scanpy
[leiden]) (8.4.0)
Requirement already satisfied: networkx>=2.7.1 in .\anaconda3\Lib\site-packages (from
scanpy[leiden]) (3.1)
Requirement already satisfied: numba!=0.62.0rc1,>=0.57.1 in .\anaconda3\Lib\site-pack
ages (from scanpy[leiden]) (0.64.0)
Requirement already satisfied: packaging>=21.3 in .\anaconda3\Lib\site-packages (from
scanpy[leiden]) (26.0)
Requirement already satisfied: pandas>=1.5.3 in .\anaconda3\Lib\site-packages (from s
canpy[leiden]) (2.3.3)
Requirement already satisfied: patsy!=1.0.0 in .\anaconda3\Lib\site-packages (from sc
anpy[leiden]) (1.0.2)
Requirement already satisfied: pynndescent>=0.5.13 in .\anaconda3\Lib\site-packages
(from scanpy[leiden]) (0.6.0)
Requirement already satisfied: scikit-learn>=1.1.3 in .\anaconda3\Lib\site-packages
(from scanpy[leiden]) (1.8.0)
Requirement already satisfied: scipy>=1.8.1 in .\anaconda3\Lib\site-packages (from sc
anpy[leiden]) (1.17.1)
Requirement already satisfied: seaborn>=0.13.2 in .\anaconda3\Lib\site-packages (from
scanpy[leiden]) (0.13.2)
Requirement already satisfied: session-info2 in .\anaconda3\Lib\site-packages (from s
canpy[leiden]) (0.4)
Requirement already satisfied: statsmodels>=0.14.5 in .\anaconda3\Lib\site-packages
(from scanpy[leiden]) (0.14.6)
Requirement already satisfied: tqdm in .\anaconda3\Lib\site-packages (from scanpy[lei
den]) (4.65.0)
Requirement already satisfied: typing-extensions in .\anaconda3\Lib\site-packages (fr
om scanpy[leiden]) (4.15.0)
Requirement already satisfied: umap-learn>=0.5.6 in .\anaconda3\Lib\site-packages (fr
om scanpy[leiden]) (0.5.11)
Collecting igraph>=0.10.8 (from scanpy[leiden])
  Downloading igraph-1.0.0-cp39-abi3-win_amd64.whl.metadata (4.5 kB)
Collecting leidenalg>=0.9.0 (from scanpy[leiden])
  Downloading leidenalg-0.11.0-cp38-abi3-win_amd64.whl.metadata (10 kB)
Requirement already satisfied: array-api-compat>=1.7.1 in .\anaconda3\Lib\site-packag
es (from anndata>=0.8->scanpy[leiden]) (1.14.0)
```

```
Requirement already satisfied: python-dateutil>=2.8.2 in .\anaconda3\Lib\site-package
s (from pandas>=1.5.3->scanpy[leiden]) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in .\anaconda3\Lib\site-packages (from pa
ndas>=1.5.3->scanpy[leiden]) (2022.7)
Requirement already satisfied: tzdata>=2022.7 in .\anaconda3\Lib\site-packages (from
pandas>=1.5.3->scanpy[leiden]) (2025.3)
Collecting texttable>=1.6.2 (from igraph>=0.10.8->scanpy[leiden])
  Downloading texttable-1.7.0-py2.py3-none-any.whl.metadata (9.8 kB)
Requirement already satisfied: contourpy>=1.0.1 in .\anaconda3\Lib\site-packages (fro
m matplotlib>=3.7.5->scanpy[leiden]) (1.0.5)
Requirement already satisfied: cycler>=0.10 in .\anaconda3\Lib\site-packages (from ma
tplotlib>=3.7.5->scanpy[leiden]) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in .\anaconda3\Lib\site-packages (fr
om matplotlib>=3.7.5->scanpy[leiden]) (4.25.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\anaconda3\Lib\site-packages (fr
om matplotlib>=3.7.5->scanpy[leiden]) (1.4.4)
Requirement already satisfied: pillow>=8 in .\anaconda3\Lib\site-packages (from matpl
otlib>=3.7.5->scanpy[leiden]) (9.4.0)
Requirement already satisfied: pyparsing>=3 in .\anaconda3\Lib\site-packages (from ma
tplotlib>=3.7.5->scanpy[leiden]) (3.0.9)
Requirement already satisfied: llvmlite<0.47,>=0.46.0dev0 in .\anaconda3\Lib\site-pac
kages (from numba!=0.62.0rc1,>=0.57.1->scanpy[leiden]) (0.46.0)
Requirement already satisfied: six>=1.5 in .\anaconda3\Lib\site-packages (from python
-dateutil>=2.8.2->pandas>=1.5.3->scanpy[leiden]) (1.16.0)
Requirement already satisfied: threadpoolctl>=3.2.0 in .\anaconda3\Lib\site-packages
(from scikit-learn>=1.1.3->scanpy[leiden]) (3.6.0)
Requirement already satisfied: wrapt<3,>=1.10 in .\anaconda3\Lib\site-packages (from
deprecated->numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0->zarr<3) (1.14.1)
Requirement already satisfied: colorama in .\anaconda3\Lib\site-packages (from tqdm->
scanpy[leiden]) (0.4.6)
Downloading numpy-1.26.4-cp311-cp311-win_amd64.whl (15.8 MB)
   ---------------------------------------- 0.0/15.8 MB ? eta -:--:--
   ------------- -------------------------- 5.2/15.8 MB 31.9 MB/s eta 0:00:01
   ---------------------------------------  15.5/15.8 MB 42.2 MB/s eta 0:00:01
   ---------------------------------------- 15.8/15.8 MB 41.5 MB/s  0:00:00
Downloading h5py-3.11.0-cp311-cp311-win_amd64.whl (3.0 MB)
   ---------------------------------------- 0.0/3.0 MB ? eta -:--:--
   ---------------------------------------- 3.0/3.0 MB 57.8 MB/s  0:00:00
Downloading zarr-2.18.7-py3-none-any.whl (211 kB)
Downloading scanpy-1.11.5-py3-none-any.whl (2.1 MB)
   ---------------------------------------- 0.0/2.1 MB ? eta -:--:--
   ---------------------------------------- 2.1/2.1 MB 29.1 MB/s  0:00:00
Downloading anndata-0.12.10-py3-none-any.whl (176 kB)
Downloading igraph-1.0.0-cp39-abi3-win_amd64.whl (3.2 MB)
   ---------------------------------------- 0.0/3.2 MB ? eta -:--:--
   ---------------------------------------- 3.2/3.2 MB 63.0 MB/s  0:00:00
Downloading leidenalg-0.11.0-cp38-abi3-win_amd64.whl (2.0 MB)
   ---------------------------------------- 0.0/2.0 MB ? eta -:--:--
   ---------------------------------------- 2.0/2.0 MB 55.7 MB/s  0:00:00
Downloading texttable-1.7.0-py2.py3-none-any.whl (10 kB)
Installing collected packages: texttable, numpy, igraph, leidenalg, h5py, zarr, annda
ta, scanpy

   ----- ---------------------------------- 1/8 [numpy]
   ----- ---------------------------------- 1/8 [numpy]
   ----- ---------------------------------- 1/8 [numpy]
   ----- ---------------------------------- 1/8 [numpy]
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   ----- ---------------------------------- 1/8 [numpy]
```

```
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   ----- ---------- ------------------------------ 2/8 [igraph]
   ----- ---------- ------------------------------ 2/8 [igraph]
   ----- ---------- ------------------------------ 2/8 [igraph]
   ----- ---------- ------------------------------ 2/8 [igraph]
   ----- ---------- ------------------------------ 2/8 [igraph]
   ----- --------------- ------------------------- 3/8 [leidenalg]
   ----- -------------------- ------------------- 4/8 [h5py]
   ----- -------------------- ------------------- 4/8 [h5py]
   ----- -------------------- ------------------- 4/8 [h5py]
   ----- -------------------- ------------------- 4/8 [h5py]
   ----- -------------------- ------------------- 4/8 [h5py]
   ----- ------------------------- -------------- 5/8 [zarr]
   ----- ------------------------- -------------- 5/8 [zarr]
   ----- ------------------------- -------------- 5/8 [zarr]
   ----- ------------------------- -------------- 5/8 [zarr]
   ----- ------------------------------ --------- 6/8 [anndata]
   ----- ------------------------------ --------- 6/8 [anndata]
   ----- ------------------------------ --------- 6/8 [anndata]
   ----- ------------------------------------ ---- 7/8 [scanpy]
   ----- ------------------------------------ ---- 7/8 [scanpy]
   ----- ------------------------------------ ---- 7/8 [scanpy]
   ----- ------------------------------------ ---- 7/8 [scanpy]
   ----- ------------------------------------ ---- 7/8 [scanpy]
   ----- ------------------------------------ ---- 7/8 [scanpy]
   ----- ---------------------------------------- 8/8 [scanpy]

Successfully installed anndata-0.12.10 h5py-3.11.0 igraph-1.0.0 leidenalg-0.11.0 nump
y-1.26.4 scanpy-1.11.5 texttable-1.7.0 zarr-2.18.7
```

```
ERROR: pip's dependency resolver does not currently take into account all the package
s that are installed. This behaviour is the source of the following dependency confli
cts.
gensim 4.3.0 requires FuzzyTM>=0.4.0, which is not installed.
tables 3.8.0 requires blosc2~=2.0.0, which is not installed.
transformers 2.1.1 requires sentencepiece, which is not installed.
```

In [1]:

```
import sys
import numpy
import h5py
import zarr
import scanpy as sc

print("Python:", sys.executable)
print("numpy:", numpy.__version__)
print("h5py:", h5py.__version__)
print("zarr:", zarr.__version__)
print("scanpy:", sc.__version__)
print("All working ✅")
```

```
C:\Users\dgous\anaconda3\Lib\site-packages\pandas\core\arrays\masked.py:61: UserWarni
ng: Pandas requires version '1.3.6' or newer of 'bottleneck' (version '1.3.5' current
ly installed).
  from pandas.core import (
Python: C:\Users\dgous\anaconda3\python.exe
numpy: 1.26.4
h5py: 3.11.0
zarr: 2.18.7
scanpy: 1.11.5
All working ✅
C:\Users\dgous\AppData\Local\Temp\ipykernel_37008\970950898.py:11: FutureWarning: `__
version__` is deprecated, use `importlib.metadata.version('scanpy')` instead
  print("scanpy:", sc.__version__)
```

In [9]:

```
import pandas as pd
import scanpy as sc

file_path = r"C:\Users\dgous\Downloads\GSE118389_counts_rsem.txt"

data = pd.read_csv(file_path, sep="\t", index_col=0)

print("Shape:", data.shape)
print(data.iloc[:5, :5])
```

```
Shape: (21785, 1534)
          PT089_P1_A01  PT089_P1_A02  PT089_P1_A03  PT089_P1_A04  PT089_P1_A05
A1BG               0.0           0.0           0.0           0.0           0.0
A1BG-AS1           0.0           0.0           0.0           0.0           0.0
A1CF               0.0           0.0           0.0           1.0           0.0
A2M                0.0           0.0           0.0         339.0           0.0
A2M-AS1            0.0           0.0           0.0           0.0           0.0
```

In [10]:

```
import pandas as pd
import scanpy as sc

file_path = r"C:/Users/dgous/downloads/GSE118389_counts_rsem.txt.gz"

data = pd.read_csv(file_path, sep="\t", index_col=0)
```

```
print("Shape of raw data:", data.shape)
print(data.iloc[:5, :5])
```

```
Shape of raw data: (21785, 1534)
          PT089_P1_A01  PT089_P1_A02  PT089_P1_A03  PT089_P1_A04  PT089_P1_A05
A1BG               0.0           0.0           0.0           0.0           0.0
A1BG-AS1           0.0           0.0           0.0           0.0           0.0
A1CF               0.0           0.0           0.0           1.0           0.0
A2M                0.0           0.0           0.0         339.0           0.0
A2M-AS1            0.0           0.0           0.0           0.0           0.0
```

In [11]:

```
import scanpy as sc

adata = sc.AnnData(data.T)

adata.var_names_make_unique()

print(adata)
```

```
AnnData object with n_obs × n_vars = 1534 × 21785
```

In [12]:

```
sc.pp.calculate_qc_metrics(adata, inplace=True)

print(adata.obs.head())
```

```
              n_genes_by_counts  log1p_n_genes_by_counts  total_counts  \
PT089_P1_A01               3556                 8.176673    2110883.35
PT089_P1_A02               3372                 8.123558    1945635.61
PT089_P1_A03               2578                 7.855157    1524090.59
PT089_P1_A04               2260                 7.723562     427386.53
PT089_P1_A05               2659                 7.886081    1511151.40

              log1p_total_counts  pct_counts_in_top_50_genes  \
PT089_P1_A01           14.562618                   32.593736
PT089_P1_A02           14.481100                   26.053517
PT089_P1_A03           14.236909                   31.257428
PT089_P1_A04           12.965446                   37.910427
PT089_P1_A05           14.228383                   35.416555

              pct_counts_in_top_100_genes  pct_counts_in_top_200_genes  \
PT089_P1_A01                    41.263051                    51.344467
PT089_P1_A02                    35.660447                    48.100413
PT089_P1_A03                    42.106772                    57.527703
PT089_P1_A04                    47.330944                    58.396803
PT089_P1_A05                    46.790784                    60.948319

              pct_counts_in_top_500_genes
PT089_P1_A01                    69.569811
PT089_P1_A02                    71.365522
PT089_P1_A03                    85.098990
PT089_P1_A04                    77.907144
PT089_P1_A05                    85.976003
```

In [13]:

```
sc.pp.filter_cells(adata, min_genes=200)
sc.pp.filter_genes(adata, min_cells=3)

print("After filtering:", adata.shape)
```

```
After filtering: (1207, 17942)
```

In [14]:
```
sc.pp.normalize_total(adata, target_sum=1e4)
sc.pp.log1p(adata)
```

In [15]:
```
sc.pp.highly_variable_genes(adata, n_top_genes=2000)

adata = adata[:, adata.var.highly_variable].copy()

print("After HVG:", adata.shape)
```

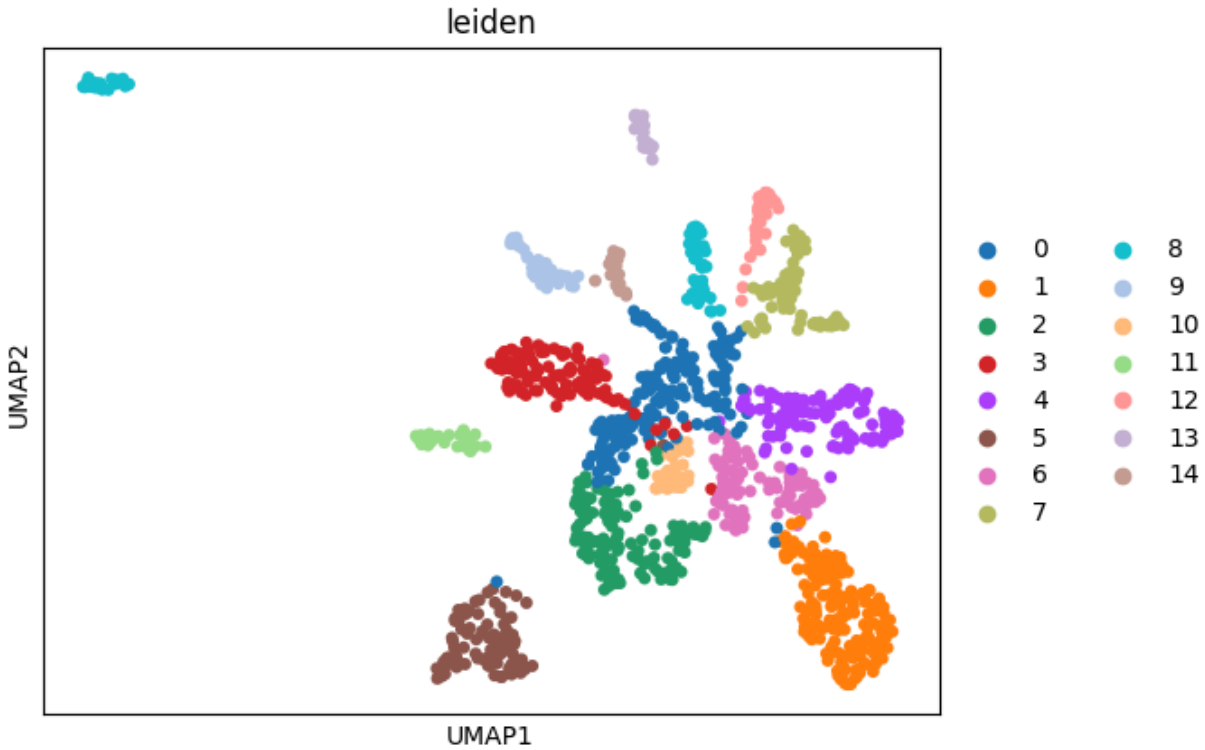
```
After HVG: (1207, 2000)
```

In [16]:
```
sc.pp.scale(adata, max_value=10)
sc.tl.pca(adata)
```

In [17]:
```
sc.pp.neighbors(adata, n_neighbors=10, n_pcs=20)
sc.tl.umap(adata)

sc.pl.umap(adata)
```

In [18]:
```
sc.tl.leiden(adata, resolution=0.5)

sc.pl.umap(adata, color="leiden")
```

```
C:\Users\dgous\AppData\Local\Temp\ipykernel_37008\506050691.py:1: FutureWarning: In t
he future, the default backend for leiden will be igraph instead of leidenalg.

 To achieve the future defaults please pass: flavor="igraph" and n_iterations=2.  dir
ected must also be False to work with igraph's implementation.
  sc.tl.leiden(adata, resolution=0.5)
```



In [19]:
```
sc.tl.rank_genes_groups(adata, "leiden", method="wilcoxon")

sc.pl.rank_genes_groups(adata, n_genes=10, sharey=False)
```

```
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
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C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
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C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
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C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
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C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
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C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:484: Ru
ntimeWarning: invalid value encountered in log2
  self.stats[group_name, "logfoldchanges"] = np.log2(
```
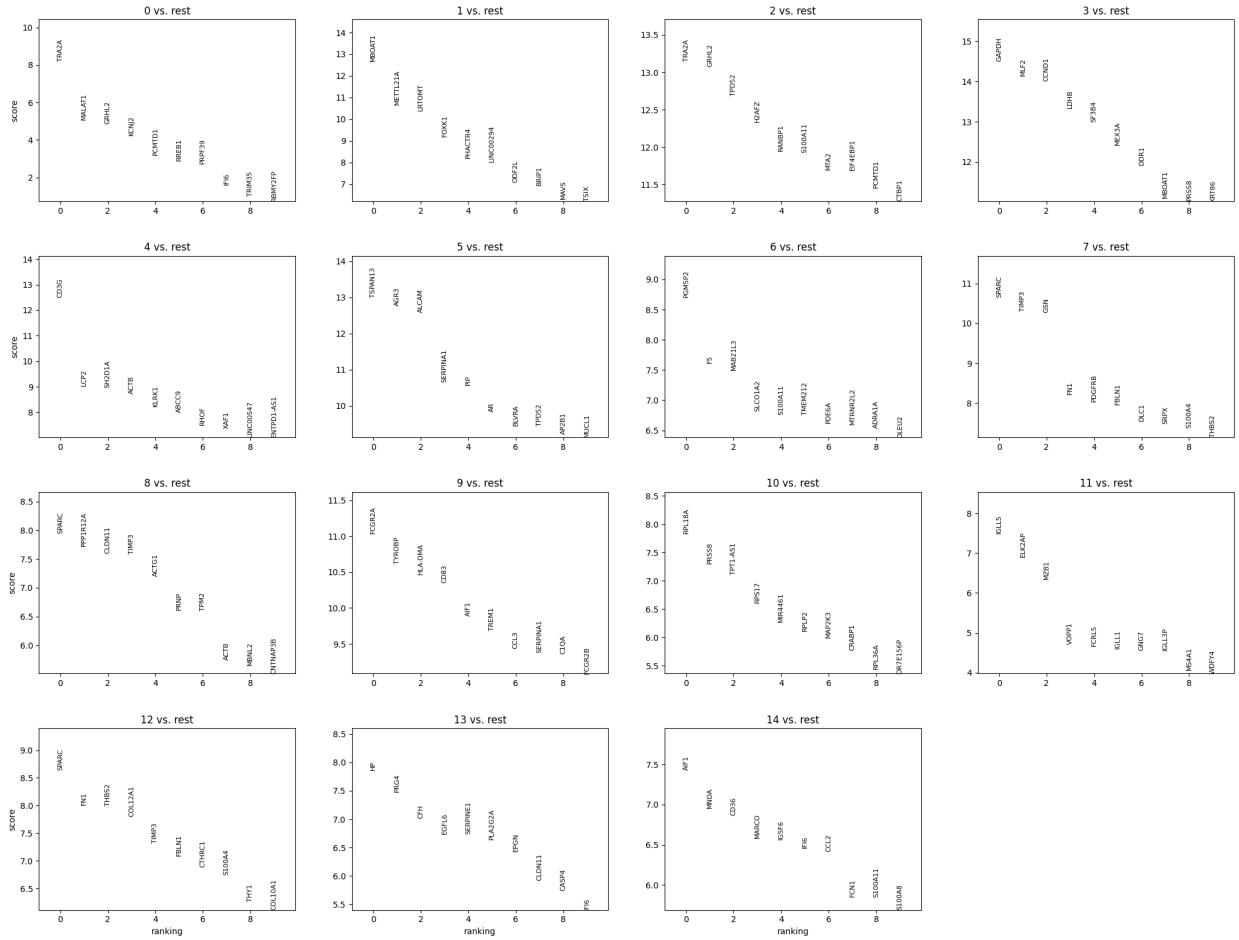
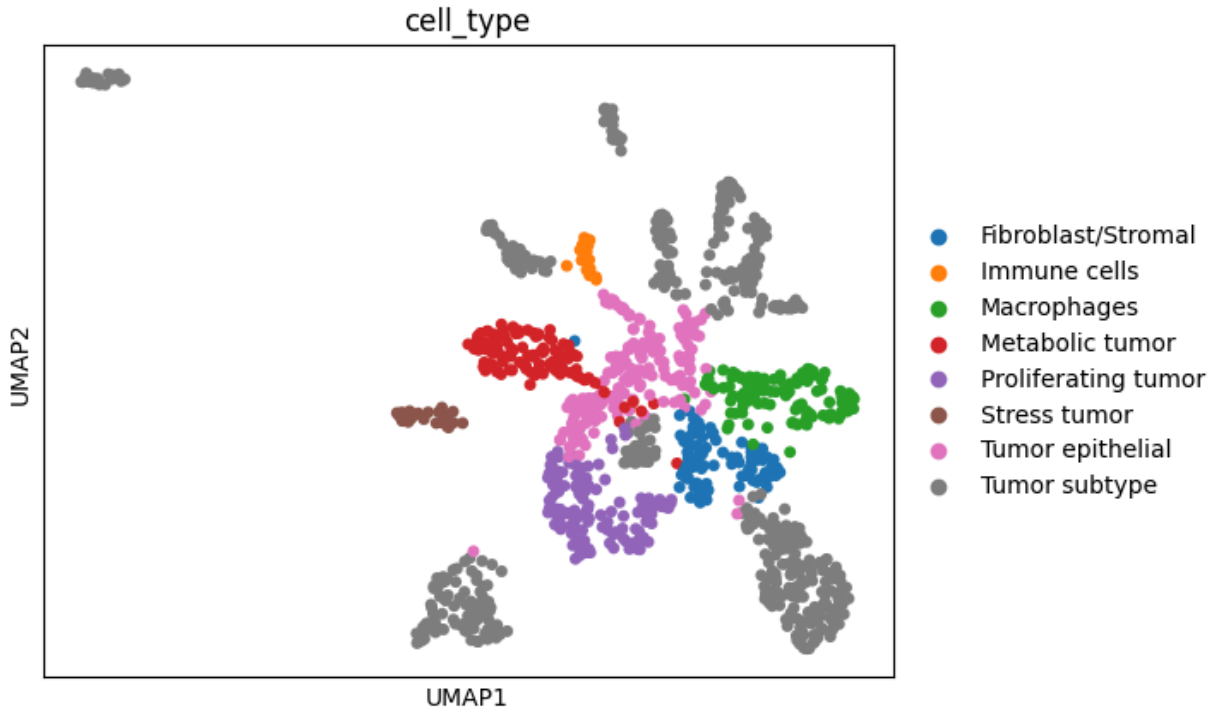



In [21]:

```
cluster_map = {
    "0": "Tumor epithelial",
    "1": "Tumor subtype",
    "2": "Proliferating tumor",
    "3": "Metabolic tumor",
    "4": "Macrophages",
    "5": "Tumor subtype",
    "6": "Fibroblast/Stromal",
    "7": "Tumor subtype",
    "8": "Tumor subtype",
    "9": "Tumor subtype",
    "10": "Tumor subtype",
    "11": "Stress tumor",
    "12": "Tumor subtype",
    "13": "Tumor subtype",
    "14": "Immune cells"
}

adata.obs["cell_type"] = adata.obs["leiden"].map(cluster_map)

sc.pl.umap(adata, color="cell_type")
```



In [22]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split

X = adata.X
y = adata.obs["cell_type"]

import pandas as pd
X_df = pd.DataFrame(X)

X_train, X_test, y_train, y_test = train_test_split(X_df, y, test_size=0.2)

model = RandomForestClassifier()
model.fit(X_train, y_train)

print("Accuracy:", model.score(X_test, y_test))
```

```
Accuracy: 0.8099173553719008
```

In [23]:
```
import numpy as np

importances = model.feature_importances_
top_idx = np.argsort(importances)[-20:]

print(adata.var_names[top_idx])
```

```
Index(['ZAP70', 'CMAS', 'RANBP1', 'FABP7', 'METTL21A', 'GSN', 'MLF2', 'PCMTD1',
       'TPD52', 'H2AFZ', 'MBOAT1', 'ODF2L', 'GAPDH', 'SH2D1A', 'CD3G',
       'S100A11', 'LDHB', 'KLRK1', 'GRHL2', 'TRA2A'],
      dtype='object')
```

In [26]:
```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

print("Accuracy:", model.score(X_test, y_test))
```

```
Accuracy: 0.9090909090909091
```

In [27]:
```python
from sklearn.svm import SVC

model = SVC(kernel='linear')
model.fit(X_train, y_train)

print("Accuracy:", model.score(X_test, y_test))
```

```
Accuracy: 0.9297520661157025
```

In [28]:
```python
!pip install xgboost
```

```
Requirement already satisfied: xgboost in .\anaconda3\Lib\site-packages (2.0.3)
Requirement already satisfied: numpy in .\anaconda3\Lib\site-packages (from xgboost)
(1.26.4)
Requirement already satisfied: scipy in .\anaconda3\Lib\site-packages (from xgboost)
(1.17.1)
```

In [30]:
```python
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from xgboost import XGBClassifier
import pandas as pd

# Features
X = pd.DataFrame(adata.X)

# Labels
y = adata.obs["cell_type"].astype(str)

# Convert text labels to numbers
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

print("Class mapping:")
for i, label in enumerate(label_encoder.classes_):
    print(i, "->", label)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y_encoded, test_size=0.2, random_state=42, stratify=y_encoded
)

# XGBoost model
model = XGBClassifier(
    objective="multi:softprob",
    num_class=len(label_encoder.classes_),
    eval_metric="mlogloss",
    random_state=42
)

model.fit(X_train, y_train)

print("Accuracy:", model.score(X_test, y_test))
```

```
Class mapping:
0 -> Fibroblast/Stromal
1 -> Immune cells
2 -> Macrophages
3 -> Metabolic tumor
4 -> Proliferating tumor
5 -> Stress tumor
6 -> Tumor epithelial
7 -> Tumor subtype
Accuracy: 0.8553719008264463
```

In [31]:

```
y_pred = model.predict(X_test)
predicted_labels = label_encoder.inverse_transform(y_pred)

print(predicted_labels[:10])
```

```
['Metabolic tumor' 'Tumor subtype' 'Tumor subtype' 'Macrophages'
 'Proliferating tumor' 'Proliferating tumor' 'Tumor subtype'
 'Tumor subtype' 'Metabolic tumor' 'Tumor subtype']
```

In [32]:

```
model = XGBClassifier(
    objective="multi:softprob",
    num_class=len(label_encoder.classes_),
    eval_metric="mlogloss",
    n_estimators=200,
    max_depth=6,
    learning_rate=0.1,
    subsample=0.8,
    colsample_bytree=0.8,
    random_state=42
)
```

In [33]:

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score

models = {
    "Random Forest": RandomForestClassifier(random_state=42),
    "Logistic Regression": LogisticRegression(max_iter=1000, random_state=42),
    "SVM": SVC(kernel="linear", random_state=42),
    "XGBoost": XGBClassifier(
        objective="multi:softprob",
        num_class=len(label_encoder.classes_),
        eval_metric="mlogloss",
        random_state=42
    )
}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    print(name, "Accuracy:", accuracy_score(y_test, y_pred))
```

```
Random Forest Accuracy: 0.8140495867768595
Logistic Regression Accuracy: 0.9338842975206612
SVM Accuracy: 0.9462809917355371
XGBoost Accuracy: 0.8553719008264463
```

In [34]:
```
from sklearn.neural_network import MLPClassifier

model = MLPClassifier(hidden_layer_sizes=(100, 50), max_iter=300)
model.fit(X_train, y_train)

print("Accuracy:", model.score(X_test, y_test))
```

```
Accuracy: 0.9380165289256198
```

In [35]:
```
from sklearn.decomposition import PCA

pca = PCA(n_components=50)
X_pca = pca.fit_transform(X_df)

X_train, X_test, y_train, y_test = train_test_split(X_pca, y, test_size=0.2)

model = MLPClassifier()
model.fit(X_train, y_train)

print("Accuracy:", model.score(X_test, y_test))
```

```
Accuracy: 0.9586776859504132
```

In [36]:
```
sc.pp.neighbors(adata)
```

In [37]:
```
sc.tl.leiden(adata)
```

In [38]:
```
!pip install scvi-tools
```

```
Collecting scvi-tools
  Downloading scvi_tools-1.4.2-py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: anndata in .\anaconda3\Lib\site-packages (from scvi-to
ols) (0.12.10)
Collecting docrep>=0.3.2 (from scvi-tools)
  Downloading docrep-0.3.2.tar.gz (33 kB)
  Installing build dependencies: started
  Installing build dependencies: finished with status 'done'
  Getting requirements to build wheel: started
  Getting requirements to build wheel: finished with status 'done'
  Preparing metadata (pyproject.toml): started
  Preparing metadata (pyproject.toml): finished with status 'done'
Collecting lightning>=2.0 (from scvi-tools)
  Downloading lightning-2.6.1-py3-none-any.whl.metadata (44 kB)
Collecting ml-collections (from scvi-tools)
  Downloading ml_collections-1.1.0-py3-none-any.whl.metadata (22 kB)
Collecting mudata (from scvi-tools)
  Downloading mudata-0.3.3-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: numba>=0.60.0 in .\anaconda3\Lib\site-packages (from s
cvi-tools) (0.64.0)
Requirement already satisfied: numpy in .\anaconda3\Lib\site-packages (from scvi-tool
s) (1.26.4)
Requirement already satisfied: pandas in .\anaconda3\Lib\site-packages (from scvi-too
ls) (2.3.3)
Collecting pyro-ppl (from scvi-tools)
  Downloading pyro_ppl-1.9.1-py3-none-any.whl.metadata (7.8 kB)
Requirement already satisfied: rich in .\anaconda3\Lib\site-packages (from scvi-tool
s) (13.7.1)
Requirement already satisfied: scanpy>=1.10 in .\anaconda3\Lib\site-packages (from sc
anpy[skmisc]>=1.10->scvi-tools) (1.11.5)
Requirement already satisfied: scikit-learn in .\anaconda3\Lib\site-packages (from sc
vi-tools) (1.8.0)
Requirement already satisfied: scipy in .\anaconda3\Lib\site-packages (from scvi-tool
s) (1.17.1)
Collecting sparse>=0.14.0 (from scvi-tools)
  Downloading sparse-0.18.0-py2.py3-none-any.whl.metadata (5.3 kB)
Requirement already satisfied: tensorboard in .\anaconda3\Lib\site-packages (from scv
i-tools) (2.16.2)
Requirement already satisfied: torch in .\anaconda3\Lib\site-packages (from scvi-tool
s) (2.2.1)
Collecting torchmetrics (from scvi-tools)
  Downloading torchmetrics-1.9.0-py3-none-any.whl.metadata (23 kB)
Requirement already satisfied: tqdm in .\anaconda3\Lib\site-packages (from scvi-tool
s) (4.65.0)
Requirement already satisfied: xarray in .\anaconda3\Lib\site-packages (from scvi-too
ls) (2023.6.0)
Requirement already satisfied: six in .\anaconda3\Lib\site-packages (from docrep>=0.
3.2->scvi-tools) (1.16.0)
Requirement already satisfied: PyYAML<8.0,>5.4 in .\anaconda3\Lib\site-packages (from
lightning>=2.0->scvi-tools) (6.0)
Requirement already satisfied: fsspec<2028.0,>=2022.5.0 in .\anaconda3\Lib\site-packa
ges (from fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tools) (2023.3.0)
Collecting lightning-utilities<2.0,>=0.10.0 (from lightning>=2.0->scvi-tools)
  Downloading lightning_utilities-0.15.3-py3-none-any.whl.metadata (5.5 kB)
Requirement already satisfied: packaging<27.0,>=23.0 in .\anaconda3\Lib\site-packages
(from lightning>=2.0->scvi-tools) (26.0)
Requirement already satisfied: typing-extensions<6.0,>4.5.0 in .\anaconda3\Lib\site-p
ackages (from lightning>=2.0->scvi-tools) (4.15.0)
Collecting pytorch-lightning (from lightning>=2.0->scvi-tools)
  Downloading pytorch_lightning-2.6.1-py3-none-any.whl.metadata (21 kB)
```

```
Requirement already satisfied: requests in .\anaconda3\Lib\site-packages (from fsspec
[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tools) (2.31.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in .\anaconda3\Lib\site-pac
kages (from fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tools) (3.8.3)
Requirement already satisfied: filelock in .\anaconda3\Lib\site-packages (from torch-
>scvi-tools) (3.9.0)
Requirement already satisfied: sympy in .\anaconda3\Lib\site-packages (from torch->sc
vi-tools) (1.11.1)
Requirement already satisfied: networkx in .\anaconda3\Lib\site-packages (from torch-
>scvi-tools) (3.1)
Requirement already satisfied: jinja2 in .\anaconda3\Lib\site-packages (from torch->s
cvi-tools) (3.1.2)
Requirement already satisfied: colorama in .\anaconda3\Lib\site-packages (from tqdm->
scvi-tools) (0.4.6)
Requirement already satisfied: attrs>=17.3.0 in .\anaconda3\Lib\site-packages (from a
iohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tool
s) (22.1.0)
Requirement already satisfied: charset-normalizer<3.0,>=2.0 in .\anaconda3\Lib\site-p
ackages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=
2.0->scvi-tools) (2.0.4)
Requirement already satisfied: multidict<7.0,>=4.5 in .\anaconda3\Lib\site-packages
(from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scv
i-tools) (6.0.2)
Requirement already satisfied: async-timeout<5.0,>=4.0.0a3 in .\anaconda3\Lib\site-pa
ckages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=
2.0->scvi-tools) (4.0.2)
Requirement already satisfied: yarl<2.0,>=1.0 in .\anaconda3\Lib\site-packages (from
aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tool
s) (1.8.1)
Requirement already satisfied: frozenlist>=1.1.1 in .\anaconda3\Lib\site-packages (fr
om aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-t
ools) (1.3.3)
Requirement already satisfied: aiosignal>=1.1.2 in .\anaconda3\Lib\site-packages (fro
m aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-to
ols) (1.2.0)
Requirement already satisfied: idna>=2.0 in .\anaconda3\Lib\site-packages (from yarl<
2.0,>=1.0->aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0
->scvi-tools) (3.4)
Requirement already satisfied: llvmlite<0.47,>=0.46.0dev0 in .\anaconda3\Lib\site-pac
kages (from numba>=0.60.0->scvi-tools) (0.46.0)
Requirement already satisfied: h5py>=3.7.0 in .\anaconda3\Lib\site-packages (from sca
npy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (3.11.0)
Requirement already satisfied: joblib in .\anaconda3\Lib\site-packages (from scanpy>=
1.10->scanpy[skmisc]>=1.10->scvi-tools) (1.5.3)
Requirement already satisfied: legacy-api-wrap>=1.4.1 in .\anaconda3\Lib\site-package
s (from scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (1.5)
Requirement already satisfied: matplotlib>=3.7.5 in .\anaconda3\Lib\site-packages (fr
om scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (3.10.8)
Requirement already satisfied: natsort in .\anaconda3\Lib\site-packages (from scanpy>
=1.10->scanpy[skmisc]>=1.10->scvi-tools) (8.4.0)
Requirement already satisfied: patsy!=1.0.0 in .\anaconda3\Lib\site-packages (from sc
anpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (1.0.2)
Requirement already satisfied: pynndescent>=0.5.13 in .\anaconda3\Lib\site-packages
(from scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (0.6.0)
Requirement already satisfied: seaborn>=0.13.2 in .\anaconda3\Lib\site-packages (from
scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (0.13.2)
Requirement already satisfied: session-info2 in .\anaconda3\Lib\site-packages (from s
canpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (0.4)
Requirement already satisfied: statsmodels>=0.14.5 in .\anaconda3\Lib\site-packages
(from scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (0.14.6)
```

```
Requirement already satisfied: umap-learn>=0.5.6 in .\anaconda3\Lib\site-packages (fr
om scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (0.5.11)
Requirement already satisfied: array-api-compat>=1.7.1 in .\anaconda3\Lib\site-packag
es (from anndata->scvi-tools) (1.14.0)
Requirement already satisfied: zarr!=3.0.*,>=2.18.7 in .\anaconda3\Lib\site-packages
(from anndata->scvi-tools) (2.18.7)
Requirement already satisfied: python-dateutil>=2.8.2 in .\anaconda3\Lib\site-package
s (from pandas->scvi-tools) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in .\anaconda3\Lib\site-packages (from pa
ndas->scvi-tools) (2022.7)
Requirement already satisfied: tzdata>=2022.7 in .\anaconda3\Lib\site-packages (from
pandas->scvi-tools) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in .\anaconda3\Lib\site-packages (fro
m matplotlib>=3.7.5->scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (1.0.5)
Requirement already satisfied: cycler>=0.10 in .\anaconda3\Lib\site-packages (from ma
tplotlib>=3.7.5->scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in .\anaconda3\Lib\site-packages (fr
om matplotlib>=3.7.5->scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (4.25.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\anaconda3\Lib\site-packages (fr
om matplotlib>=3.7.5->scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (1.4.4)
Requirement already satisfied: pillow>=8 in .\anaconda3\Lib\site-packages (from matpl
otlib>=3.7.5->scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (9.4.0)
Requirement already satisfied: pyparsing>=3 in .\anaconda3\Lib\site-packages (from ma
tplotlib>=3.7.5->scanpy>=1.10->scanpy[skmisc]>=1.10->scvi-tools) (3.0.9)
Collecting scikit-misc>=0.1.4 (from scanpy[skmisc]>=1.10->scvi-tools)
  Downloading scikit_misc-0.5.2-cp311-cp311-win_amd64.whl.metadata (5.0 kB)
Requirement already satisfied: threadpoolctl>=3.2.0 in .\anaconda3\Lib\site-packages
(from scikit-learn->scvi-tools) (3.6.0)
Requirement already satisfied: asciitree in .\anaconda3\Lib\site-packages (from zarr!
=3.0.*,>=2.18.7->anndata->scvi-tools) (0.3.3)
Requirement already satisfied: fasteners in .\anaconda3\Lib\site-packages (from zarr!
=3.0.*,>=2.18.7->anndata->scvi-tools) (0.20)
Requirement already satisfied: numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0 in .\anacond
a3\Lib\site-packages (from zarr!=3.0.*,>=2.18.7->anndata->scvi-tools) (0.15.1)
Requirement already satisfied: deprecated in .\anaconda3\Lib\site-packages (from numc
odecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0->zarr!=3.0.*,>=2.18.7->anndata->scvi-tools) (1.
3.1)
Requirement already satisfied: wrapt<3,>=1.10 in .\anaconda3\Lib\site-packages (from
deprecated->numcodecs!=0.14.0,!=0.14.1,<0.16,>=0.10.0->zarr!=3.0.*,>=2.18.7->anndata-
>scvi-tools) (1.14.1)
Requirement already satisfied: MarkupSafe>=2.0 in .\anaconda3\Lib\site-packages (from
jinja2->torch->scvi-tools) (2.1.1)
Requirement already satisfied: absl-py in .\anaconda3\Lib\site-packages (from ml-coll
ections->scvi-tools) (2.1.0)
Requirement already satisfied: opt-einsum>=2.3.2 in .\anaconda3\Lib\site-packages (fr
om pyro-ppl->scvi-tools) (3.3.0)
Collecting pyro-api>=0.1.1 (from pyro-ppl->scvi-tools)
  Downloading pyro_api-0.1.2-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: urllib3<3,>=1.21.1 in .\anaconda3\Lib\site-packages (f
rom requests->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tools) (1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in .\anaconda3\Lib\site-packages (f
rom requests->fsspec[http]<2028.0,>=2022.5.0->lightning>=2.0->scvi-tools) (2023.7.22)
Requirement already satisfied: markdown-it-py>=2.2.0 in .\anaconda3\Lib\site-packages
(from rich->scvi-tools) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in .\anaconda3\Lib\site-packag
es (from rich->scvi-tools) (2.15.1)
Requirement already satisfied: mdurl~=0.1 in .\anaconda3\Lib\site-packages (from mark
down-it-py>=2.2.0->rich->scvi-tools) (0.1.0)
Requirement already satisfied: mpmath>=0.19 in .\anaconda3\Lib\site-packages (from sy
mpy->torch->scvi-tools) (1.3.0)
```

```
Requirement already satisfied: grpcio>=1.48.2 in .\anaconda3\Lib\site-packages (from
tensorboard->scvi-tools) (1.62.1)
Requirement already satisfied: markdown>=2.6.8 in .\anaconda3\Lib\site-packages (from
tensorboard->scvi-tools) (3.4.1)
Requirement already satisfied: protobuf!=4.24.0,>=3.19.6 in .\anaconda3\Lib\site-pack
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rom tensorboard->scvi-tools) (68.0.0)
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ib\site-packages (from tensorboard->scvi-tools) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in .\anaconda3\Lib\site-packages (from
tensorboard->scvi-tools) (2.2.3)
Downloading scvi_tools-1.4.2-py3-none-any.whl (662 kB)
   ---------------------------------------- 0.0/662.1 kB ? eta -:--:--
   ---------------------------------------- 662.1/662.1 kB 6.5 MB/s  0:00:00
Downloading lightning-2.6.1-py3-none-any.whl (853 kB)
   ---------------------------------------- 0.0/853.6 kB ? eta -:--:--
   ---------------------------------------- 853.6/853.6 kB 12.6 MB/s  0:00:00
Downloading lightning_utilities-0.15.3-py3-none-any.whl (31 kB)
Downloading torchmetrics-1.9.0-py3-none-any.whl (983 kB)
   ---------------------------------------- 0.0/983.4 kB ? eta -:--:--
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Downloading scikit_misc-0.5.2-cp311-cp311-win_amd64.whl (152 kB)
Downloading sparse-0.18.0-py2.py3-none-any.whl (151 kB)
Downloading ml_collections-1.1.0-py3-none-any.whl (76 kB)
Downloading mudata-0.3.3-py3-none-any.whl (43 kB)
Downloading pyro_ppl-1.9.1-py3-none-any.whl (755 kB)
   ---------------------------------------- 0.0/756.0 kB ? eta -:--:--
   ---------------------------------------- 756.0/756.0 kB 15.9 MB/s  0:00:00
Downloading pyro_api-0.1.2-py3-none-any.whl (11 kB)
Downloading pytorch_lightning-2.6.1-py3-none-any.whl (857 kB)
   ---------------------------------------- 0.0/857.3 kB ? eta -:--:--
   ---------------------------------------- 857.3/857.3 kB 39.7 MB/s  0:00:00
Building wheels for collected packages: docrep
  Building wheel for docrep (pyproject.toml): started
  Building wheel for docrep (pyproject.toml): finished with status 'done'
  Created wheel for docrep: filename=docrep-0.3.2-py3-none-any.whl size=19998 sha256=
9fff3083eb5f640764fc91d808e744c07a8c315ef6d4748836d75852002f1b67
  Stored in directory: c:\users\dgous\appdata\local\pip\cache\wheels\06\76\8f\0ecb7d3
57c0bff71a2bd1940671be2d07a200752da9189bb55
Successfully built docrep
Installing collected packages: pyro-api, scikit-misc, ml-collections, lightning-utili
ties, docrep, sparse, torchmetrics, pyro-ppl, pytorch-lightning, mudata, lightning, s
cvi-tools

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```
Successfully installed docrep-0.3.2 lightning-2.6.1 lightning-utilities-0.15.3 ml-col
lections-1.1.0 mudata-0.3.3 pyro-api-0.1.2 pyro-ppl-1.9.1 pytorch-lightning-2.6.1 sci
kit-misc-0.5.2 scvi-tools-1.4.2 sparse-0.18.0 torchmetrics-1.9.0
```

In [46]:

```python
import pandas as pd
import scanpy as sc
import scvi
import numpy as np

file_path = r"C:\Users\dgous\Downloads\GSE118389_counts_rsem.txt.gz"
raw_counts = pd.read_csv(file_path, sep="\t", index_col=0)

adata_scvi = sc.AnnData(raw_counts.T)
adata_scvi.var_names_make_unique()
adata_scvi.X = np.asarray(adata_scvi.X, dtype=np.float32)

sc.pp.filter_cells(adata_scvi, min_genes=200)
sc.pp.filter_genes(adata_scvi, min_cells=3)

print(adata_scvi)
print("NaN:", np.isnan(adata_scvi.X).any())
print("Inf:", np.isinf(adata_scvi.X).any())
```

```
AnnData object with n_obs × n_vars = 1207 × 17942
    obs: 'n_genes'
    var: 'n_cells'
NaN: False
Inf: False
```

In [50]:

```python
scvi.model.SCVI.setup_anndata(adata_scvi)

model_scvi = scvi.model.SCVI(
    adata_scvi,
    n_latent=10,
    n_layers=2,
    n_hidden=128,
    gene_likelihood="nb"
)

model_scvi.train(
    max_epochs=50,
    early_stopping=True
)

adata_scvi.obsm["X_scVI"] = model_scvi.get_latent_representation()
print(adata_scvi.obsm["X_scVI"].shape)
```

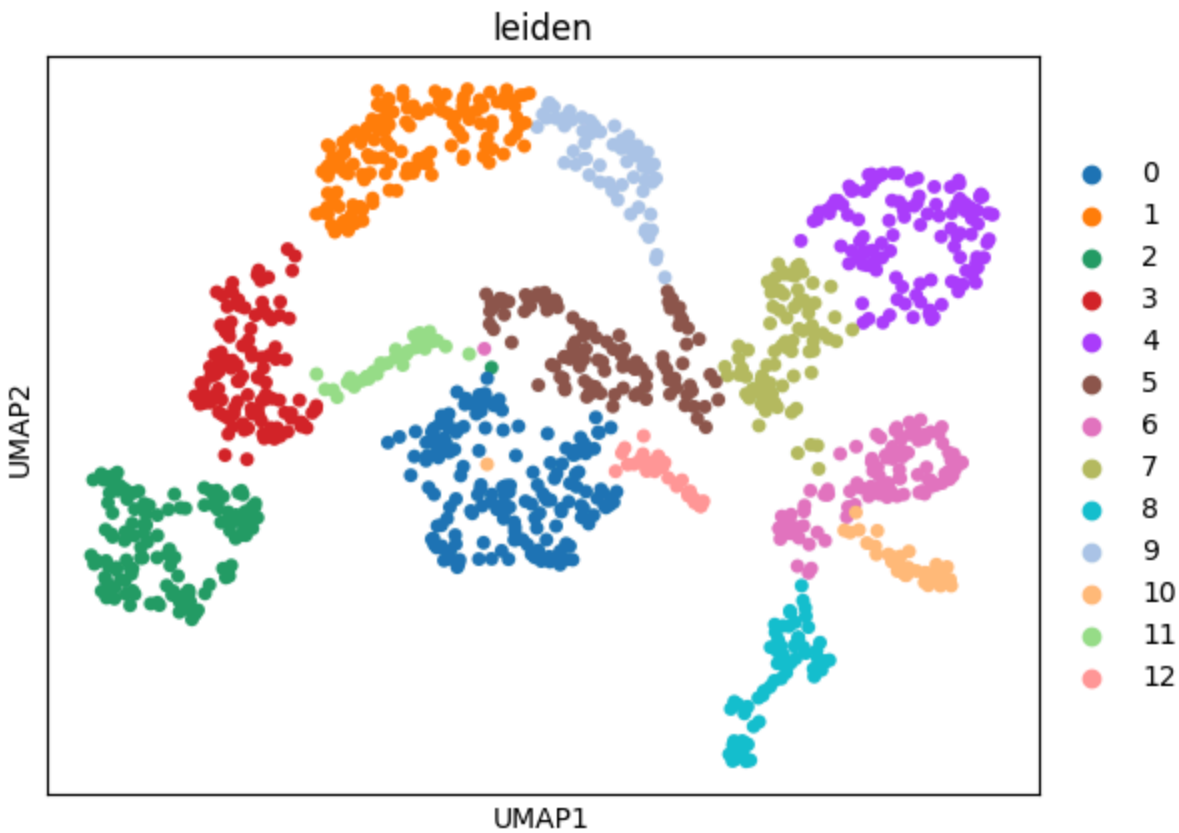
```
C:\Users\dgous\anaconda3\Lib\site-packages\scvi\data\fields\_base_field.py:63: UserWa
rning: adata.X does not contain unnormalized count data. Are you sure this is what yo
u want?
  self.validate_field(adata)
GPU available: False, used: False
TPU available: False, using: 0 TPU cores
💡 Tip: For seamless cloud logging and experiment tracking, try installing [litlogge
r](https://pypi.org/project/litlogger/) to enable LitLogger, which logs metrics and a
rtifacts automatically to the Lightning Experiments platform.
C:\Users\dgous\anaconda3\Lib\site-packages\lightning\pytorch\trainer\connectors\data_
connector.py:434: The 'train_dataloader' does not have many workers which may be a bo
ttleneck. Consider increasing the value of the `num_workers` argument` to `num_worker
s=11` in the `DataLoader` to improve performance.
C:\Users\dgous\anaconda3\Lib\site-packages\lightning\pytorch\loops\fit_loop.py:317: T
he number of training batches (9) is smaller than the logging interval Trainer(log_ev
ery_n_steps=10). Set a lower value for log_every_n_steps if you want to see logs for
the training epoch.
C:\Users\dgous\anaconda3\Lib\site-packages\lightning\pytorch\trainer\connectors\data_
connector.py:434: The 'val_dataloader' does not have many workers which may be a bott
leneck. Consider increasing the value of the `num_workers` argument` to `num_workers=
11` in the `DataLoader` to improve performance.
Training:   0%|          | 0/50 [00:00<?, ?it/s]
```

```
`Trainer.fit` stopped: `max_epochs=50` reached.
(1207, 10)
```

In [51]:

```python
sc.pp.neighbors(adata_scvi, use_rep="X_scVI")
sc.tl.umap(adata_scvi)
sc.tl.leiden(adata_scvi, resolution=0.5)

sc.pl.umap(adata_scvi, color="leiden")
```



In [52]:
```
sc.tl.rank_genes_groups(adata_scvi, "leiden", method="wilcoxon")

sc.pl.rank_genes_groups(adata_scvi, n_genes=10, sharey=False)
```

```
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:481: Ru
ntimeWarning: overflow encountered in expm1
  foldchanges = (self.expm1_func(mean_group) + 1e-9) / (
C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:482: Ru
ntimeWarning: overflow encountered in expm1
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C:\Users\dgous\anaconda3\Lib\site-packages\scanpy\tools\_rank_genes_groups.py:481: Ru
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```
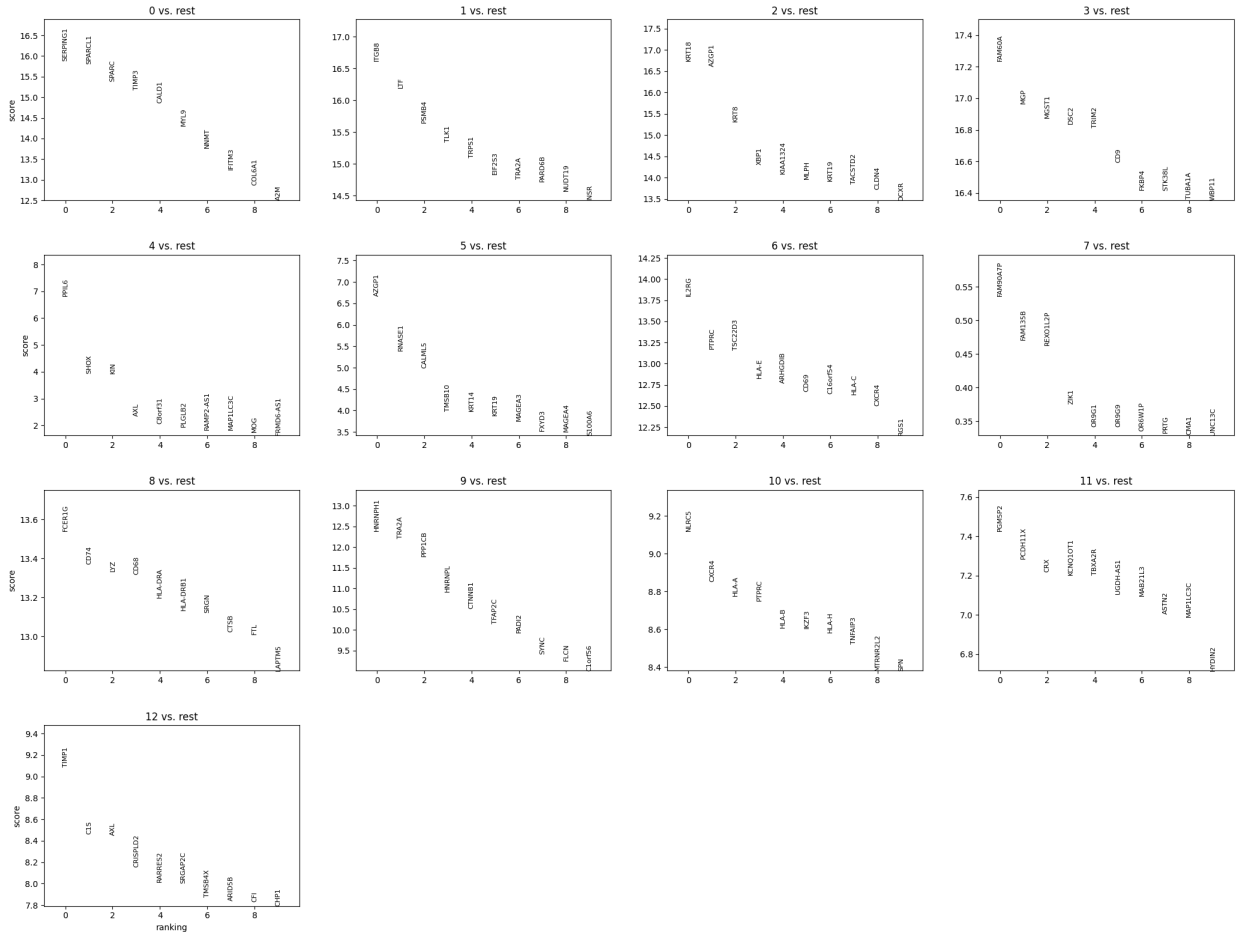
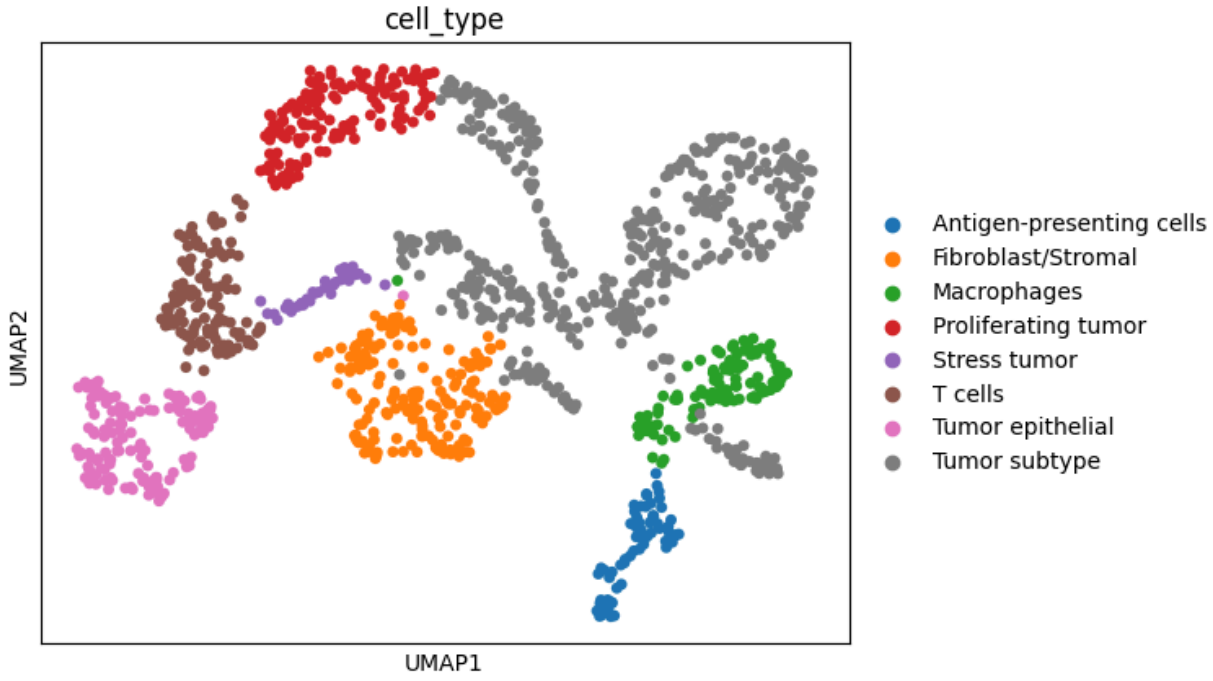



In [53]:

```
cluster_map_scvi = {
    "0": "Fibroblast/Stromal",
    "1": "Proliferating tumor",
    "2": "Tumor epithelial",
    "3": "T cells",
    "4": "Tumor subtype",
    "5": "Tumor subtype",
    "6": "Macrophages",
    "7": "Tumor subtype",
    "8": "Antigen-presenting cells",
    "9": "Tumor subtype",
    "10": "Tumor subtype",
    "11": "Stress tumor",
    "12": "Tumor subtype"
}

adata_scvi.obs["cell_type"] = adata_scvi.obs["leiden"].map(cluster_map_scvi)

sc.pl.umap(adata_scvi, color="cell_type")
```



In [54]:
```
markers = sc.get.rank_genes_groups_df(adata_scvi, group=None)
markers.head(20)
```

Out[54]:

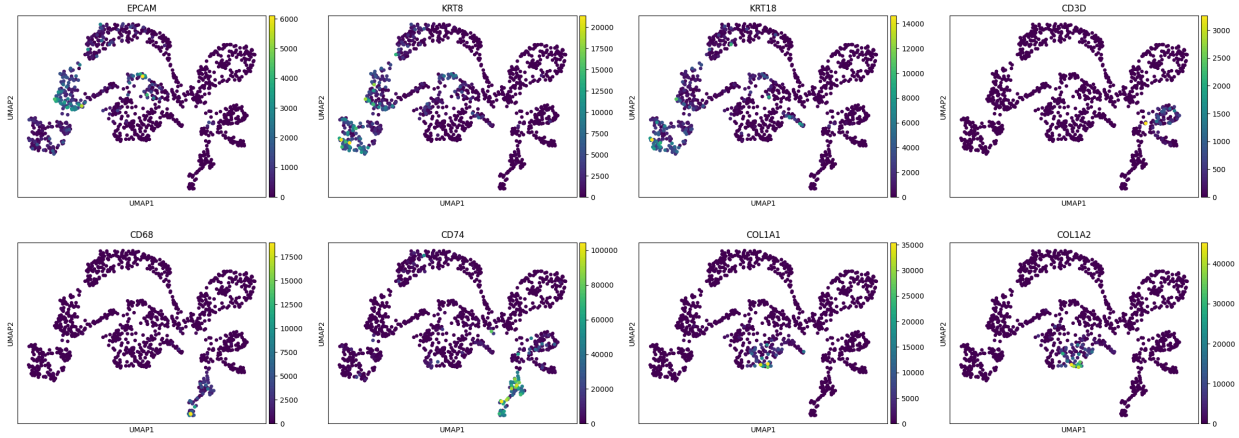
	group	names	scores	logfoldchanges	pvals	pvals_adj
0	0	SERPING1	15.875337	inf	9.390560e-57	1.684854e-52
1	0	SPARCL1	15.808915	inf	2.700794e-56	2.422882e-52
2	0	SPARC	15.381848	inf	2.166729e-53	1.295848e-49
3	0	TIMP3	15.177017	inf	5.020983e-52	2.252162e-48
4	0	CALD1	14.863259	inf	5.707928e-50	2.048233e-46
5	0	MYL9	14.308558	125.270676	1.934824e-46	5.785769e-43
6	0	NNMT	13.762501	inf	4.284461e-43	1.098168e-39
7	0	IFITM3	13.243320	644.993713	4.931966e-40	1.106117e-36
8	0	COL6A1	12.860535	459.003723	7.505567e-38	1.496276e-34
9	0	A2M	12.499416	inf	7.520097e-36	1.349256e-32
10	0	EGR1	12.392621	NaN	2.865247e-35	4.673479e-32
11	0	MMP2	12.223664	inf	2.323684e-34	3.474295e-31
12	0	C1S	12.207208	inf	2.844876e-34	3.926366e-31
13	0	TAGLN	12.150257	inf	5.718544e-34	7.328723e-31
14	0	RBMS3	12.134628	337.498688	6.922350e-34	8.280053e-31
15	0	C1R	12.128590	inf	7.452142e-34	8.356646e-31
16	0	FSTL1	12.089992	inf	1.192985e-33	1.259090e-30
17	0	IFI16	12.002495	483.619781	3.447456e-33	3.436347e-30
18	0	MT2A	11.961529	223.709396	5.651192e-33	5.336510e-30
19	0	THBS1	11.896527	inf	1.233749e-32	1.106797e-29

In [55]:

```
markers.to_csv("scvi_cluster_markers.csv", index=False)
```

In [56]:

```
sc.pl.umap(
    adata_scvi,
    color=["EPCAM", "KRT8", "KRT18", "CD3D", "CD68", "CD74", "COL1A1", "COL1A2"],
    ncols=4
)
```



In [57]:
```
print(adata_scvi.obs["leiden"].value_counts().sort_index())
print(adata_scvi.obs["cell_type"].value_counts())
```

```
leiden
0     171
1     139
2     136
3     120
4     118
5     108
6     103
7      82
8      66
9      60
10     38
11     35
12     31
Name: count, dtype: int64
cell_type
Tumor subtype               437
Fibroblast/Stromal          171
Proliferating tumor         139
Tumor epithelial            136
T cells                     120
Macrophages                 103
Antigen-presenting cells     66
Stress tumor                 35
Name: count, dtype: int64
```

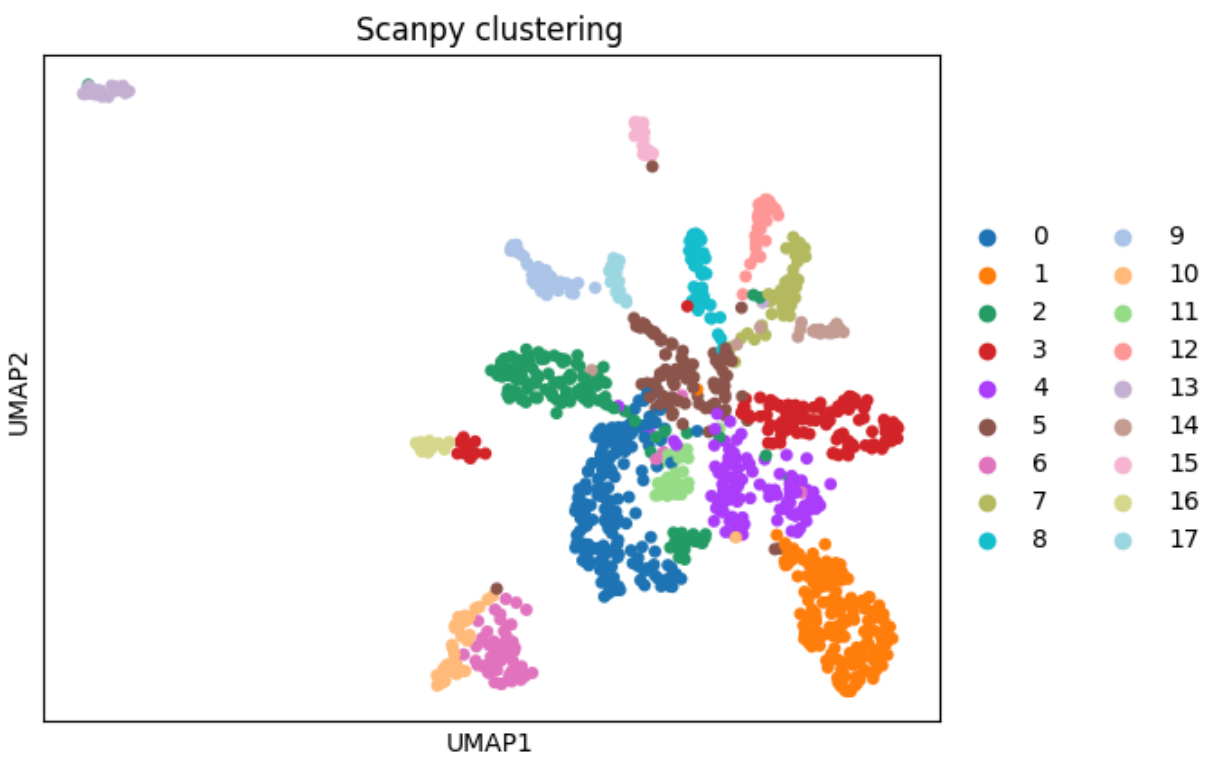
In [58]:
```
summary = pd.DataFrame({
    "cluster": sorted(adata_scvi.obs["leiden"].unique()),
})

summary["cell_count"] = summary["cluster"].map(
    adata_scvi.obs["leiden"].value_counts()
)

summary["assigned_cell_type"] = summary["cluster"].map(cluster_map_scvi)

summary
```

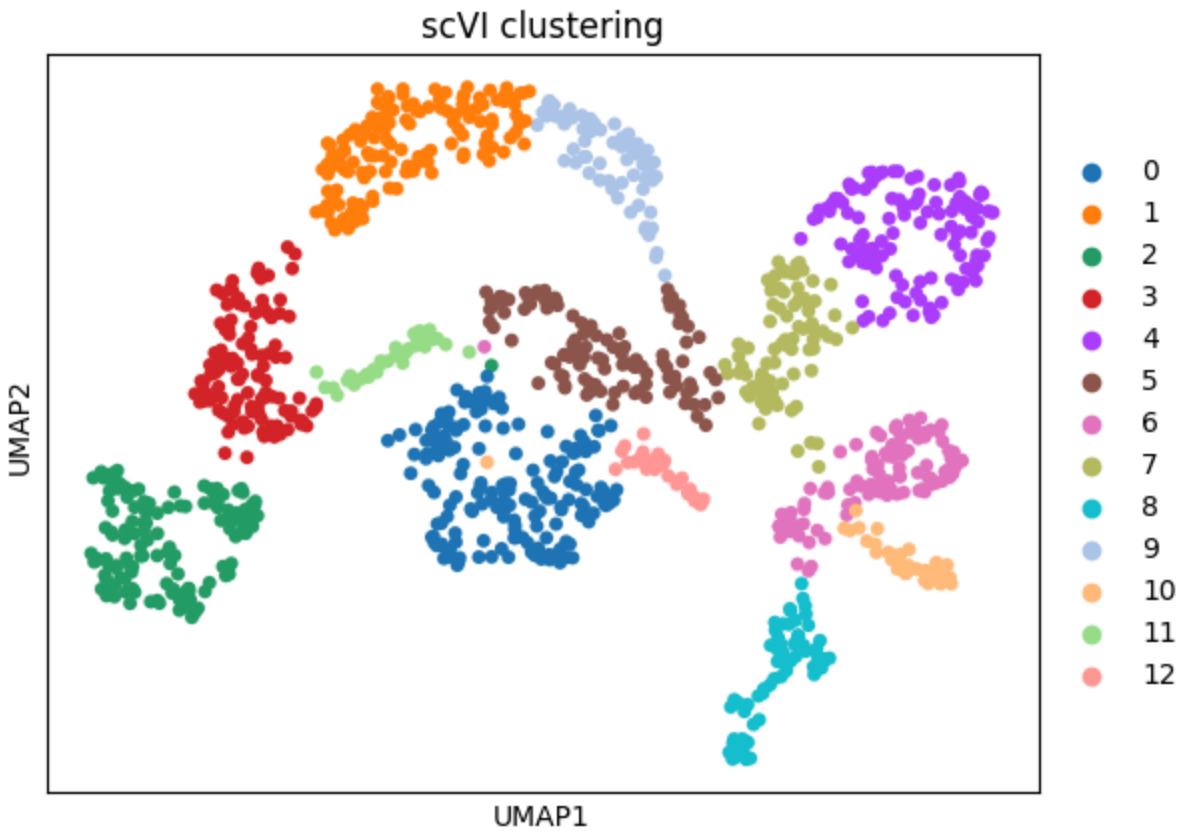
Out[58]:

	cluster	cell_count	assigned_cell_type
0	0	171	Fibroblast/Stromal
1	1	139	Proliferating tumor
2	10	38	Tumor subtype
3	11	35	Stress tumor
4	12	31	Tumor subtype
5	2	136	Tumor epithelial
6	3	120	T cells
7	4	118	Tumor subtype
8	5	108	Tumor subtype
9	6	103	Macrophages
10	7	82	Tumor subtype
11	8	66	Antigen-presenting cells
12	9	60	Tumor subtype

In [59]:
```
summary.to_csv("scvi_cluster_summary.csv", index=False)
```

In [60]:
```
sc.pl.umap(adata, color="leiden", title="Scanpy clustering")
sc.pl.umap(adata_scvi, color="leiden", title="scVI clustering")
```





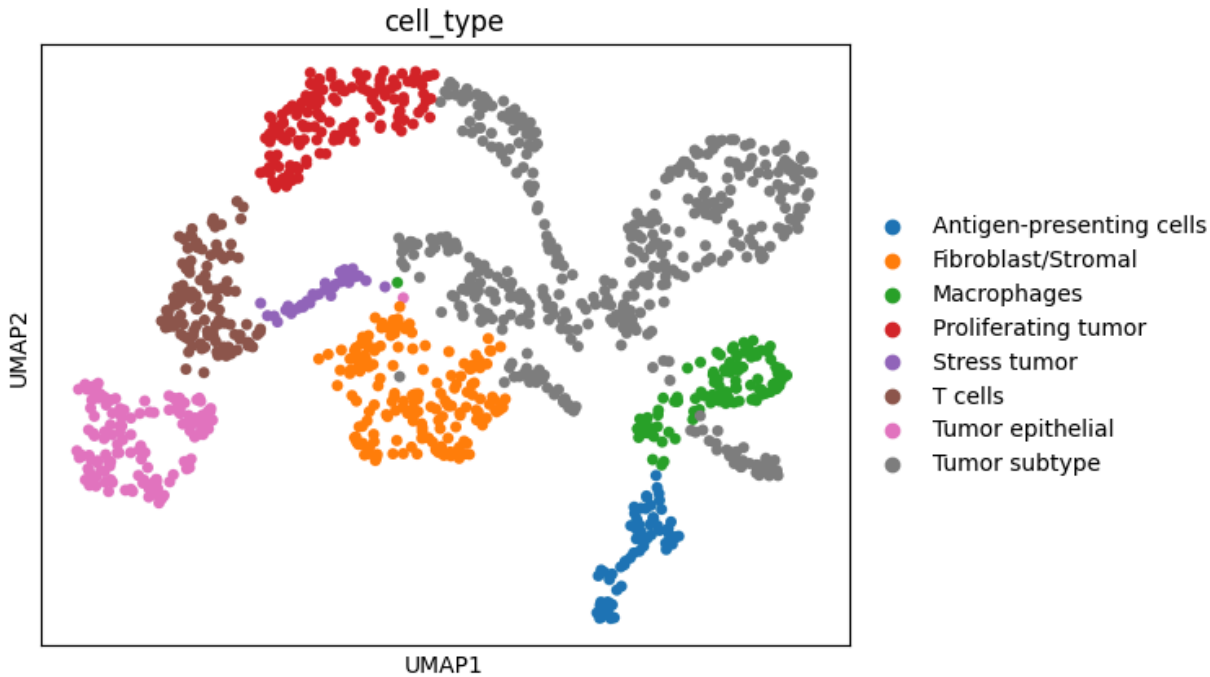
In [61]:

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, accuracy_score
import pandas as pd

X = pd.DataFrame(adata_scvi.obsm["X_scVI"])
y = adata_scvi.obs["cell_type"].astype(str)

le = LabelEncoder()
y_enc = le.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X, y_enc, test_size=0.2, random_state=42, stratify=y_enc
)

rf = RandomForestClassifier(random_state=42)
rf.fit(X_train, y_train)

y_pred = rf.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred, target_names=le.classes_))
```

```
Accuracy: 0.9834710743801653
                          precision    recall  f1-score   support

Antigen-presenting cells       1.00      1.00      1.00        13
      Fibroblast/Stromal       1.00      1.00      1.00        34
             Macrophages       0.95      0.95      0.95        21
     Proliferating tumor       1.00      1.00      1.00        28
            Stress tumor       0.88      1.00      0.93         7
                 T cells       1.00      0.96      0.98        24
        Tumor epithelial       1.00      0.96      0.98        27
           Tumor subtype       0.98      0.99      0.98        88

                accuracy                           0.98       242
               macro avg       0.98      0.98      0.98       242
            weighted avg       0.98      0.98      0.98       242
```

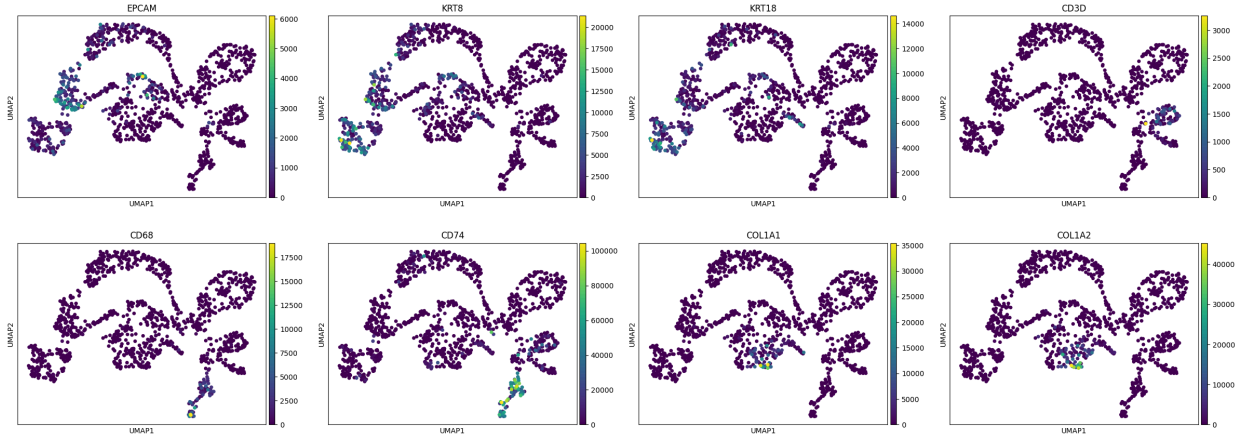
In [62]:

```
adata_scvi.obs["cell_type"] = adata_scvi.obs["leiden"].map(cluster_map_scvi)
sc.pl.umap(adata_scvi, color="cell_type")
```



In [63]:

```
sc.pl.umap(
    adata_scvi,
    color=["EPCAM", "KRT8", "KRT18", "CD3D", "CD68", "CD74", "COL1A1", "COL1A2"],
    ncols=4
)
```



In [64]:
```
markers = sc.get.rank_genes_groups_df(adata_scvi, group=None)
markers.to_csv("scvi_cluster_markers.csv", index=False)
```

In [65]:
```
import pandas as pd
import scanpy as sc

# Full marker table
markers = sc.get.rank_genes_groups_df(adata_scvi, group=None)

# Save all markers
markers.to_csv("scvi_all_markers.csv", index=False)

# Keep only significant/top markers
top_markers = markers[
    (markers["pvals_adj"] < 0.05) & (markers["logfoldchanges"] > 1)
].copy()

top_markers.to_csv("scvi_significant_markers.csv", index=False)

print(top_markers.head())
print("Number of significant markers:", top_markers.shape[0])
```

```
  group     names     scores  logfoldchanges         pvals     pvals_adj
0     0  SERPING1  15.875337             inf  9.390560e-57  1.684854e-52
1     0   SPARCL1  15.808915             inf  2.700794e-56  2.422882e-52
2     0     SPARC  15.381848             inf  2.166729e-53  1.295848e-49
3     0     TIMP3  15.177017             inf  5.020983e-52  2.252162e-48
4     0     CALD1  14.863259             inf  5.707928e-50  2.048233e-46
Number of significant markers: 20095
```

In [66]:
```
top10_per_cluster = (
    top_markers.sort_values(["group", "scores"], ascending=[True, False])
    .groupby("group")
    .head(10)
)

top10_per_cluster.to_csv("scvi_top10_markers_per_cluster.csv", index=False)

print(top10_per_cluster[["group", "names", "scores", "logfoldchanges"]].head(30))
```

```
      group     names     scores  logfoldchanges
0         0  SERPING1  15.875337             inf
1         0   SPARCL1  15.808915             inf
2         0     SPARC  15.381848             inf
3         0     TIMP3  15.177017             inf
4         0     CALD1  14.863259             inf
5         0      MYL9  14.308558      125.270676
6         0      NNMT  13.762501             inf
7         0    IFITM3  13.243320      644.993713
8         0    COL6A1  12.860535      459.003723
9         0       A2M  12.499416             inf
17942     1     ITGB8  16.616100             inf
17943     1       LTF  16.188501             inf
17944     1     PSMB4  15.642555             inf
17945     1      TLK1  15.346883             inf
17946     1     TRPS1  15.097644             inf
17948     1     TRA2A  14.757091             inf
17949     1    PARD6B  14.721910             inf
17950     1    NUDT19  14.572264             inf
17951     1      INSR  14.428049             inf
17952     1    TFAP2C  14.408389             inf
35884     2     KRT18  16.739958             inf
35885     2     AZGP1  16.613823             inf
35886     2      KRT8  15.310805             inf
35887     2      XBP1  14.306155             inf
35888     2  KIAA1324  14.090705             inf
35889     2      MLPH  13.962740      614.469116
35891     2   TACSTD2  13.848617             inf
35892     2     CLDN4  13.718172             inf
35893     2      DCXR  13.470991      284.180786
35894     2  SERPINA3  13.348381             inf
```

```
C:\Users\dgous\AppData\Local\Temp\ipykernel_37008\1663607766.py:3: FutureWarning: The
default of observed=False is deprecated and will be changed to True in a future versi
on of pandas. Pass observed=False to retain current behavior or observed=True to adop
t the future default and silence this warning.
  .groupby("group")
```

In [67]:

```python
cluster_gene_lists = {}

for cluster in sorted(top_markers["group"].astype(str).unique(), key=lambda x: int(x))
    genes = top_markers[top_markers["group"].astype(str) == cluster]["names"].dropna()
    cluster_gene_lists[cluster] = genes[:100]  # keep top 100 genes

for cluster, genes in cluster_gene_lists.items():
    print(f"Cluster {cluster}: {genes[:10]}")
```

```
Cluster 0: ['SERPING1', 'SPARCL1', 'SPARC', 'TIMP3', 'CALD1', 'MYL9', 'NNMT', 'IFITM
3', 'COL6A1', 'A2M']
Cluster 1: ['ITGB8', 'LTF', 'PSMB4', 'TLK1', 'TRPS1', 'TRA2A', 'PARD6B', 'NUDT19', 'I
NSR', 'TFAP2C']
Cluster 2: ['KRT18', 'AZGP1', 'KRT8', 'XBP1', 'KIAA1324', 'MLPH', 'TACSTD2', 'CLDN4',
'DCXR', 'SERPINA3']
Cluster 3: ['FAM60A', 'MGST1', 'DSC2', 'TRIM2', 'CD9', 'FKBP4', 'STK38L', 'TUBA1A',
'WBP11', 'DDX47']
Cluster 4: ['PPIL6', 'SHOX', 'STX1B', 'FLG2', 'EMX2OS', 'CTAGE5', 'PTCHD4', 'FABP5',
'NANOG', 'SLC9A4']
Cluster 5: ['RNASE1', 'CALML5', 'TMSB10', 'KRT14', 'MAGEA3', 'FXYD3', 'MAGEA4', 'S100
A6', 'CDKN2A', 'IFI6']
Cluster 6: ['IL2RG', 'PTPRC', 'TSC22D3', 'HLA-E', 'ARHGDIB', 'CD69', 'C16orf54', 'HLA
-C', 'CXCR4', 'RGS1']
Cluster 7: ['MS4A2', 'UTY', 'JAKMIP2', 'SREK1IP1', 'PAG1', 'ATP6V0A2']
Cluster 8: ['FCER1G', 'LYZ', 'CD68', 'HLA-DRA', 'HLA-DRB1', 'SRGN', 'CTSB', 'LAPTM5',
'HLA-DRB5', 'MS4A7']
Cluster 9: ['TRA2A', 'HNRNPL', 'CTNNB1', 'TFAP2C', 'PADI2', 'SYNC', 'FLCN', 'C1orf5
6', 'GRHL2', 'PCMTD1']
Cluster 10: ['NLRC5', 'CXCR4', 'HLA-A', 'PTPRC', 'HLA-B', 'IKZF3', 'HLA-H', 'TNFAIP
3', 'SPN', 'TMC8']
Cluster 11: ['PGM5P2', 'PCDH11X', 'CRX', 'TBXA2R', 'UGDH-AS1', 'MAB21L3', 'ASTN2', 'M
AP1LC3C', 'HYDIN2', 'UCKL1-AS1']
Cluster 12: ['TIMP1', 'C1S', 'AXL', 'CRISPLD2', 'RARRES2', 'SRGAP2C', 'ARID5B', 'CF
I', 'CTTN', 'EIF5AL1']
```

In [68]:

```
!pip install gseapy
```

```
Collecting gseapy
  Downloading gseapy-1.1.13-cp311-cp311-win_amd64.whl.metadata (12 kB)
Requirement already satisfied: numpy>=1.13.0 in .\anaconda3\Lib\site-packages (from g
seapy) (1.26.4)
Requirement already satisfied: scipy in .\anaconda3\Lib\site-packages (from gseapy)
(1.17.1)
Requirement already satisfied: pandas in .\anaconda3\Lib\site-packages (from gseapy)
(2.3.3)
Requirement already satisfied: matplotlib>=2.2 in .\anaconda3\Lib\site-packages (from
gseapy) (3.10.8)
Requirement already satisfied: requests in .\anaconda3\Lib\site-packages (from gseap
y) (2.31.0)
Requirement already satisfied: contourpy>=1.0.1 in .\anaconda3\Lib\site-packages (fro
m matplotlib>=2.2->gseapy) (1.0.5)
Requirement already satisfied: cycler>=0.10 in .\anaconda3\Lib\site-packages (from ma
tplotlib>=2.2->gseapy) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in .\anaconda3\Lib\site-packages (fr
om matplotlib>=2.2->gseapy) (4.25.0)
Requirement already satisfied: kiwisolver>=1.3.1 in .\anaconda3\Lib\site-packages (fr
om matplotlib>=2.2->gseapy) (1.4.4)
Requirement already satisfied: packaging>=20.0 in .\anaconda3\Lib\site-packages (from
matplotlib>=2.2->gseapy) (26.0)
Requirement already satisfied: pillow>=8 in .\anaconda3\Lib\site-packages (from matpl
otlib>=2.2->gseapy) (9.4.0)
Requirement already satisfied: pyparsing>=3 in .\anaconda3\Lib\site-packages (from ma
tplotlib>=2.2->gseapy) (3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in .\anaconda3\Lib\site-packages
(from matplotlib>=2.2->gseapy) (2.8.2)
Requirement already satisfied: six>=1.5 in .\anaconda3\Lib\site-packages (from python
-dateutil>=2.7->matplotlib>=2.2->gseapy) (1.16.0)
Requirement already satisfied: pytz>=2020.1 in .\anaconda3\Lib\site-packages (from pa
ndas->gseapy) (2022.7)
Requirement already satisfied: tzdata>=2022.7 in .\anaconda3\Lib\site-packages (from
pandas->gseapy) (2025.3)
Requirement already satisfied: charset-normalizer<4,>=2 in .\anaconda3\Lib\site-packa
ges (from requests->gseapy) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in .\anaconda3\Lib\site-packages (from re
quests->gseapy) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in .\anaconda3\Lib\site-packages (f
rom requests->gseapy) (1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in .\anaconda3\Lib\site-packages (f
rom requests->gseapy) (2023.7.22)
Downloading gseapy-1.1.13-cp311-cp311-win_amd64.whl (418 kB)
Installing collected packages: gseapy
Successfully installed gseapy-1.1.13
```

In [76]:

```
print(type(cluster_gene_lists))
print(cluster_gene_lists.keys())
print(list(cluster_gene_lists.keys())[:10])
```

```
<class 'dict'>
dict_keys(['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12'])
['0', '1', '2', '3', '4', '5', '6', '7', '8', '9']
```

In [75]:

```
cluster_id = "2"
```

In [73]:

```
gene_list = [g for g in gene_list if pd.notna(g)]
```

In [78]:
```
cluster_id = "2"

if cluster_id not in cluster_gene_lists:
    cluster_id = int(cluster_id)

gene_list = cluster_gene_lists[cluster_id]
print(gene_list[:10])
```

```
['KRT18', 'AZGP1', 'KRT8', 'XBP1', 'KIAA1324', 'MLPH', 'TACSTD2', 'CLDN4', 'DCXR', 'S
ERPINA3']
```

In [79]:
```
print(list(cluster_gene_lists.keys()))
```

```
['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', '10', '11', '12']
```

In [84]:
```
import pandas as pd
import gseapy as gp

all_enrichment_results = []

for cluster_id, gene_list in cluster_gene_lists.items():
    try:
        # clean gene list
        gene_list = [str(g).upper() for g in gene_list if pd.notna(g)]
        gene_list = list(dict.fromkeys(gene_list))  # remove duplicates, keep order

        if len(gene_list) < 5:
            print(f"Skipping cluster {cluster_id}: too few genes")
            continue

        enr = gp.enrichr(
            gene_list=gene_list,
            gene_sets=[
                "KEGG_2021_Human",
                "GO_Biological_Process_2023",
                "Reactome_2022"
            ],
            organism="human",   # <- fixed
            outdir=None,
            cutoff=0.5
        )

        if enr.results is None or enr.results.empty:
            print(f"No enrichment results for cluster {cluster_id}")
            continue

        res = enr.results.copy()
        res["cluster"] = str(cluster_id)
        all_enrichment_results.append(res)

        print(f"Cluster {cluster_id} done, {res.shape[0]} pathways found")

    except Exception as e:
        print(f"Cluster {cluster_id} failed: {e}")

if len(all_enrichment_results) > 0:
    enrichment_df = pd.concat(all_enrichment_results, ignore_index=True)
    enrichment_df.to_csv("scvi_pathway_enrichment_all_clusters.csv", index=False)
    print("enrichment_df created ✅")
    print(enrichment_df.head(20))
```

```python
else:
    print("❌ No enrichment results found")
```

```
Cluster 0 done, 1592 pathways found
Cluster 1 done, 1543 pathways found
Cluster 2 done, 1296 pathways found
Cluster 3 done, 1619 pathways found
Cluster 4 done, 240 pathways found
Cluster 5 done, 1074 pathways found
Cluster 6 done, 1641 pathways found
Cluster 7 done, 56 pathways found
Cluster 8 done, 1576 pathways found
Cluster 9 done, 1543 pathways found
Cluster 10 done, 1647 pathways found
Cluster 11 done, 553 pathways found
Cluster 12 done, 1504 pathways found
enrichment_df created ✅
           Gene_set                                                Term  \
0   KEGG_2021_Human                                       Focal adhesion
1   KEGG_2021_Human                            ECM-receptor interaction
2   KEGG_2021_Human                             Proteoglycans in cancer
3   KEGG_2021_Human                          PI3K-Akt signaling pathway
4   KEGG_2021_Human                       Human papillomavirus infection
5   KEGG_2021_Human                                           Amoebiasis
6   KEGG_2021_Human                     Protein digestion and absorption
7   KEGG_2021_Human                     Regulation of actin cytoskeleton
8   KEGG_2021_Human                 Leukocyte transendothelial migration
9   KEGG_2021_Human                               Small cell lung cancer
10  KEGG_2021_Human  AGE-RAGE signaling pathway in diabetic complic...
11  KEGG_2021_Human                                  Platelet activation
12  KEGG_2021_Human                           Relaxin signaling pathway
13  KEGG_2021_Human                                           Pertussis
14  KEGG_2021_Human                  Complement and coagulation cascades
15  KEGG_2021_Human                                       Bladder cancer
16  KEGG_2021_Human                                  Pathways in cancer
17  KEGG_2021_Human                                       Axon guidance
18  KEGG_2021_Human                   Vascular smooth muscle contraction
19  KEGG_2021_Human                               p53 signaling pathway

   Overlap       P-value  Adjusted P-value  Old P-value  Old Adjusted P-value  \
0   17/201  1.757921e-16      2.109506e-14            0                     0
1    10/88  2.014446e-11      1.208667e-09            0                     0
2   13/205  3.059468e-11      1.223787e-09            0                     0
3   13/354  2.388030e-08      7.164091e-07            0                     0
4   12/331  9.878319e-08      2.370797e-06            0                     0
5    7/102  7.911542e-07      1.449250e-05            0                     0
6    7/103  8.453960e-07      1.449250e-05            0                     0
7    8/218  1.379808e-05      2.069712e-04            0                     0
8    6/114  2.316363e-05      3.088484e-04            0                     0
9     5/92  9.846889e-05      1.181627e-03            0                     0
10   5/100  1.460736e-04      1.593530e-03            0                     0
11   5/124  3.974780e-04      3.974780e-03            0                     0
12   5/129  4.764430e-04      4.397935e-03            0                     0
13    4/76  5.728566e-04      4.910199e-03            0                     0
14    4/85  8.735952e-04      6.988761e-03            0                     0
15    3/41  1.126327e-03      8.447456e-03            0                     0
16   9/531  1.370612e-03      9.674907e-03            0                     0
17   5/182  2.214133e-03      1.476088e-02            0                     0
18   4/133  4.479389e-03      2.829088e-02            0                     0
19    3/73  5.855884e-03      3.513530e-02            0                     0

    Odds Ratio  Combined Score  \
0    21.946831      796.170215
```

```
1    28.236467      695.410312
2    15.337883      371.333144
3     8.570701      150.417626
4     8.370333      135.016310
5    15.691568      220.462962
6    15.527330      217.125805
7     8.153209       91.242408
8    11.697400      124.845494
9    11.986086      110.580870
10   10.972299       96.900761
11    8.748784       68.506222
12    8.393888       64.206213
13   11.474537       85.655986
14   10.194959       71.802009
15   16.165491      109.744172
16    3.671466       24.204136
17    5.864704       35.850318
18    6.385982       34.537105
19    8.761414       45.036369

                                                 Genes cluster
0   ITGB1;ACTN1;CAV1;LAMA4;FN1;PDGFA;LAMB1;THBS1;M...       0
1   ITGB1;COL1A1;COL1A2;COL6A2;LAMA4;COL6A1;FN1;CO...       0
2   ITGB1;CD63;CDKN1A;LUM;CAV1;MMP2;FN1;THBS1;DCN;...       0
3   ITGB1;CDKN1A;LAMA4;FN1;PDGFA;LAMB1;THBS1;COL1A...       0
4   COL1A1;ITGB1;CDKN1A;CCND2;COL1A2;COL6A2;LAMA4;...       0
5          COL1A1;COL3A1;COL1A2;ACTN1;LAMA4;FN1;LAMB1       0
6   COL1A1;COL3A1;COL1A2;COL14A1;COL6A2;COL6A1;COL6A3       0
7          ITGB1;CXCL12;GSN;ACTN1;FN1;PDGFA;MYL9;MYLK       0
8                   ITGB1;CXCL12;ACTN1;MMP2;THY1;MYL9       0
9                        ITGB1;CDKN1A;LAMA4;FN1;LAMB1       0
10                      COL1A1;COL3A1;COL1A2;MMP2;FN1       0
11                    COL1A1;ITGB1;COL3A1;COL1A2;MYLK       0
12                    COL1A1;ACTA2;COL3A1;COL1A2;MMP2       0
13                             ITGB1;C1S;C1R;SERPING1       0
14                               C1S;C1R;SERPING1;A2M       0
15                                  CDKN1A;MMP2;THBS1       0
16  ITGB1;CDKN1A;CCND2;CXCL12;MMP2;LAMA4;FN1;PDGFA...       0
17                    SEMA5A;ITGB1;CXCL12;DPYSL2;MYL9       0
18                              ACTA2;CALD1;MYL9;MYLK       0
19                                 CDKN1A;CCND2;THBS1       0
```

In [85]:

```
top_pathways = (
    enrichment_df.sort_values(["cluster", "Adjusted P-value"], ascending=[True, True])
    .groupby("cluster")
    .head(5)
)

top_pathways.to_csv("scvi_top_pathways_per_cluster.csv", index=False)

print(top_pathways[["cluster", "Gene_set", "Term", "Adjusted P-value", "Genes"]].head(
```

	cluster	Gene_set	\
1260	0	Reactome_2022	
0	0	KEGG_2021_Human	
1261	0	Reactome_2022	
1262	0	Reactome_2022	
1263	0	Reactome_2022	
2654	1	Reactome_2022	
2655	1	Reactome_2022	
2656	1	Reactome_2022	
2657	1	Reactome_2022	
2658	1	Reactome_2022	
13474	10	Reactome_2022	
12180	10	KEGG_2021_Human	
12181	10	KEGG_2021_Human	
13475	10	Reactome_2022	
13476	10	Reactome_2022	
13875	11	GO_Biological_Process_2023	
13876	11	GO_Biological_Process_2023	
13877	11	GO_Biological_Process_2023	
13878	11	GO_Biological_Process_2023	
13879	11	GO_Biological_Process_2023	
15569	12	Reactome_2022	
14380	12	KEGG_2021_Human	
15570	12	Reactome_2022	
15571	12	Reactome_2022	
15572	12	Reactome_2022	
3268	2	GO_Biological_Process_2023	
4047	2	Reactome_2022	
4048	2	Reactome_2022	
4049	2	Reactome_2022	
4050	2	Reactome_2022	

	Term	Adjusted P-value	\
1260	Extracellular Matrix Organization R-HSA-1474244	1.265653e-21	
0	Focal adhesion	2.109506e-14	
1261	Platelet Degranulation R-HSA-114608	3.068689e-13	
1262	Response To Elevated Platelet Cytosolic Ca2+ R...	3.578186e-13	
1263	Regulation Of IGF Transport And Uptake By IGFB...	3.538840e-12	
2654	Post-translational Protein Modification R-HSA-...	4.806934e-04	
2655	Metabolism Of Proteins R-HSA-392499	6.710982e-04	
2656	PCP/CE Pathway R-HSA-4086400	6.710982e-04	
2657	ER-Phagosome Pathway R-HSA-1236974	6.710982e-04	
2658	Regulation Of RUNX3 Expression And Activity R-...	7.067556e-04	
13474	Immune System R-HSA-168256	3.593335e-10	
12180	Human T-cell leukemia virus 1 infection	5.742828e-10	
12181	Viral myocarditis	3.852491e-08	
13475	Immunoregulatory Interactions Between A Lympho...	1.023629e-07	
13476	Endosomal/Vacuolar Pathway R-HSA-1236977	1.465268e-07	
13875	Regulation Of Vasoconstriction (GO:0019229)	3.526344e-01	
13876	Calcium Ion Import Across Plasma Membrane (GO:...	3.526344e-01	
13877	SCF-dependent Proteasomal Ubiquitin-Dependent ...	3.526344e-01	
13878	Cell-Cell Signaling Involved In Cardiac Conduc...	3.526344e-01	
13879	Urea Cycle (GO:0000050)	3.526344e-01	
15569	Regulation Of Complement Cascade R-HSA-977606	7.723413e-07	
14380	Complement and coagulation cascades	1.318499e-06	
15570	Complement Cascade R-HSA-166658	1.657963e-06	
15571	Immune System R-HSA-168256	4.785781e-06	
15572	Interferon Alpha/Beta Signaling R-HSA-909733	1.265301e-04	
3268	Cytoplasmic Translation (GO:0002181)	7.166447e-07	
4047	Peptide Chain Elongation R-HSA-156902	2.080705e-06	

```
4048                Viral mRNA Translation R-HSA-192823      2.080705e-06
4049      Eukaryotic Translation Elongation R-HSA-156842      2.080705e-06
4050   Formation Of A Pool Of Free 40S Subunits R-HSA...      2.848370e-06

                                                   Genes
1260   ITGB1;SPARC;COL14A1;LAMA4;PDGFA;FBLN1;THBS1;EF...
0      ITGB1;ACTN1;CAV1;LAMA4;FN1;PDGFA;LAMB1;THBS1;M...
1261   LGALS3BP;CD63;SPARC;ACTN1;RARRES2;FN1;PDGFA;TH...
1262   LGALS3BP;CD63;SPARC;ACTN1;RARRES2;FN1;PDGFA;TH...
1263   IGFBP4;MMP2;FN1;LAMB1;PRSS23;FSTL1;CST3;LGALS1...
2654   RAB2A;B3GALNT2;TFAP2C;NOP58;PSMD11;UBA6;CTBP1;...
2655   PSMD11;UBA6;CTBP1;MRPL13;PSMB4;PSMD4;SUMO1;PPP...
2656                PSMB4;PSMD11;PSMD4;DAAM1;PSME2;PFN1
2657               PSMB4;PSMD11;PSMD4;PSME2;CALR;SEC22B
2658                    PSMB4;PSMD11;PSMD4;CDKN2A;PSME2
13474  ARHGAP9;WIPF1;ITGB2;NLRC5;TNFAIP3;SLC2A3;MEFV;...
12180  TGFB1;ITGB2;HLA-B;HLA-C;CD3G;HLA-A;HLA-F;CD3E;...
12181     ITGB2;RAC2;HLA-B;HLA-C;HLA-A;ITGAL;HLA-F;HLA-G
13475  CD96;ITGB2;HLA-B;HLA-C;HLA-A;HLA-F;CD3E;ITGAL;...
13476                      HLA-B;HLA-C;HLA-A;HLA-F;HLA-G
13875                                       CHRM3;TBXA2R
13876                                     SLC24A4;TRPV1
13877                                     FBXO27;FBXL18
13878                                            RNF207
13879                                         SLC25A15
15569                    C3;C1S;CFH;C1R;CFI;SERPING1;CFB
14380           C3;C1S;CFH;C1R;SERPINE1;CFI;SERPING1;CFB
15570                    C3;C1S;CFH;C1R;CFI;SERPING1;CFB
15571  IFITM3;IFITM1;IFITM2;CFH;C1S;C1R;CFI;PTEN;IFI6...
15572              IFITM3;IFITM1;IFITM2;IFI27;STAT2;IFI6
3268   RPS15;RPS14;RPL10;RPLP1;RPS5;EIF3I;RPLP0;RPL29...
4047       RPS15;RPS14;RPLP1;RPS5;RPLP0;RPL29;EEF2;RPS23
4048       RPS15;RPS14;RPLP1;RPS5;RPLP0;RPL29;GNL3;RPS23
4049       RPS15;RPS14;RPLP1;RPS5;RPLP0;RPL29;EEF2;RPS23
4050      RPS15;RPS14;RPS5;RPLP1;EIF3I;RPLP0;RPL29;RPS23
```

In [86]:

```python
cluster_id = "2"
gene_list = [str(g).upper() for g in cluster_gene_lists[cluster_id] if pd.notna(g)]

enr = gp.enrichr(
    gene_list=gene_list,
    gene_sets=["KEGG_2021_Human", "GO_Biological_Process_2023", "Reactome_2022"],
    organism="human",
    outdir=None,
    cutoff=0.5
)

print(enr.results.head(20))
```

	Gene_set	Term	Overlap	\
0	KEGG_2021_Human	Ribosome	8/158	
1	KEGG_2021_Human	Cell adhesion molecules	7/148	
2	KEGG_2021_Human	Coronavirus disease	8/232	
3	KEGG_2021_Human	Leukocyte transendothelial migration	5/114	
4	KEGG_2021_Human	Adherens junction	3/71	
5	KEGG_2021_Human	Biosynthesis of unsaturated fatty acids	2/27	
6	KEGG_2021_Human	Fatty acid elongation	2/27	
7	KEGG_2021_Human	Vasopressin-regulated water reabsorption	2/44	
8	KEGG_2021_Human	Rap1 signaling pathway	4/210	
9	KEGG_2021_Human	AMPK signaling pathway	3/120	
10	KEGG_2021_Human	Estrogen signaling pathway	3/137	
11	KEGG_2021_Human	Hedgehog signaling pathway	2/56	
12	KEGG_2021_Human	Glutathione metabolism	2/57	
13	KEGG_2021_Human	Hepatitis C	3/157	
14	KEGG_2021_Human	Tight junction	3/169	
15	KEGG_2021_Human	Thyroid hormone synthesis	2/75	
16	KEGG_2021_Human	Pathogenic Escherichia coli infection	3/197	
17	KEGG_2021_Human	Nitrogen metabolism	1/17	
18	KEGG_2021_Human	Proteoglycans in cancer	3/205	
19	KEGG_2021_Human	Prostate cancer	2/97	

	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio	\
0	0.000001	0.000170	0	0	11.449275	
1	0.000009	0.000631	0	0	10.547777	
2	0.000022	0.000958	0	0	7.638199	
3	0.000270	0.008963	0	0	9.556253	
4	0.005420	0.144174	0	0	9.020012	
5	0.008008	0.152154	0	0	16.224490	
6	0.008008	0.152154	0	0	16.224490	
7	0.020429	0.297330	0	0	9.649174	
8	0.021192	0.297330	0	0	3.983414	
9	0.022356	0.297330	0	0	5.229448	
10	0.031419	0.338198	0	0	4.562086	
11	0.031999	0.338198	0	0	7.500378	
12	0.033057	0.338198	0	0	7.363636	
13	0.044189	0.419792	0	0	3.965591	
14	0.052911	0.451094	0	0	3.676686	
15	0.054267	0.451094	0	0	5.542913	
16	0.076185	0.546630	0	0	3.141567	
17	0.081715	0.546630	0	0	12.553030	
18	0.083541	0.546630	0	0	3.015923	
19	0.084924	0.546630	0	0	4.254565	

	Combined Score	Genes
0	155.376758	RPS15;RPS14;RPL10;RPS5;RPLP1;RPLP0;RPL29;RPS23
1	121.982554	CLDN4;ALCAM;CLDN3;SDC4;CDH1;CLDN7;PTPRF
2	82.052246	RPS15;RPS14;RPL10;RPS5;RPLP1;RPLP0;RPL29;RPS23
3	78.540207	VAV3;CLDN4;CLDN3;CTNND1;CLDN7
4	47.063258	CDH1;CTNND1;PTPRF
5	78.320495	ELOVL5;TECR
6	78.320495	ELOVL5;TECR
7	37.542776	CREB3L4;RAB11A
8	15.352643	EFNA1;VAV3;CDH1;CTNND1
9	19.875444	CREB3L4;FASN;EEF2
10	15.786317	KRT18;CREB3L4;FKBP5
11	25.816583	CSNK1A1;LRP2
12	25.106497	GSTM3;IDH2
13	12.369824	CLDN4;CLDN3;CLDN7
14	10.806283	CLDN4;CLDN3;CLDN7

```
15       16.151167                                   CREB3L4;LRP2
16        8.088238                               CLDN4;CLDN3;CLDN7
17       31.439343                                           CA12
18        7.486768                                 VAV3;SDC4;PDCD4
19       10.491764                                  SPINT1;CREB3L4
```

In [87]:

```
print(enrichment_df.shape)
print(top_pathways.shape)
print(top_pathways.head(10))
```

```
(15884, 11)
(65, 11)
            Gene_set                                                Term  \
1260    Reactome_2022    Extracellular Matrix Organization R-HSA-1474244
0     KEGG_2021_Human                                     Focal adhesion
1261    Reactome_2022                Platelet Degranulation R-HSA-114608
1262    Reactome_2022  Response To Elevated Platelet Cytosolic Ca2+ R...
1263    Reactome_2022  Regulation Of IGF Transport And Uptake By IGFB...
2654    Reactome_2022  Post-translational Protein Modification R-HSA-...
2655    Reactome_2022                Metabolism Of Proteins R-HSA-392499
2656    Reactome_2022                       PCP/CE Pathway R-HSA-4086400
2657    Reactome_2022                 ER-Phagosome Pathway R-HSA-1236974
2658    Reactome_2022  Regulation Of RUNX3 Expression And Activity R-...

      Overlap       P-value  Adjusted P-value  Old P-value  \
1260   25/291  3.812207e-24      1.265653e-21            0
0      17/201  1.757921e-16      2.109506e-14            0
1261   14/125  1.848608e-15      3.068689e-13            0
1262   14/130  3.233301e-15      3.578186e-13            0
1263   13/123  4.263662e-14      3.538840e-12            0
2654  22/1383  9.993626e-07      4.806934e-04            0
2655  25/1890  4.561774e-06      6.710982e-04            0
2656     6/88  5.225599e-06      6.710982e-04            0
2657     6/89  5.580858e-06      6.710982e-04            0
2658     5/54  7.356942e-06      7.067556e-04            0

      Old Adjusted P-value  Odds Ratio  Combined Score  \
1260                     0   24.604010     1326.742549
0                        0   21.946831      796.170215
1261                     0   29.022208      984.559362
1262                     0   27.764234      926.361250
1263                     0   26.882968      827.620730
2654                     0    3.841990       53.081506
2655                     0    3.223414       39.640895
2656                     0   15.426570      187.617034
2657                     0   15.239938      184.344857
2658                     0   21.322234      252.025954

                                                 Genes cluster
1260  ITGB1;SPARC;COL14A1;LAMA4;PDGFA;FBLN1;THBS1;EF...       0
0     ITGB1;ACTN1;CAV1;LAMA4;FN1;PDGFA;LAMB1;THBS1;M...       0
1261  LGALS3BP;CD63;SPARC;ACTN1;RARRES2;FN1;PDGFA;TH...       0
1262  LGALS3BP;CD63;SPARC;ACTN1;RARRES2;FN1;PDGFA;TH...       0
1263  IGFBP4;MMP2;FN1;LAMB1;PRSS23;FSTL1;CST3;LGALS1...       0
2654  RAB2A;B3GALNT2;TFAP2C;NOP58;PSMD11;UBA6;CTBP1;...       1
2655  PSMD11;UBA6;CTBP1;MRPL13;PSMB4;PSMD4;SUMO1;PPP...       1
2656                PSMB4;PSMD11;PSMD4;DAAM1;PSME2;PFN1       1
2657               PSMB4;PSMD11;PSMD4;PSME2;CALR;SEC22B       1
2658                    PSMB4;PSMD11;PSMD4;CDKN2A;PSME2       1
```

In [88]:
```
drug_target_dict = {
    "EPCAM": ["Epithelial-targeting therapies"],
    "TACSTD2": ["TROP2-targeted therapies"],
    "CD3D": ["T-cell immunotherapy"],
    "CD68": ["Macrophage targeting (TAMs)"],
    "CD74": ["Immune modulation"],
    "COL1A1": ["Fibroblast/stroma targeting"],
    "COL1A2": ["Fibroblast/stroma targeting"],
    "KRT8": ["Epithelial tumor marker"],
    "KRT18": ["Epithelial tumor marker"],
    "MKI67": ["Cell cycle inhibitors"],
    "TOP2A": ["Topoisomerase inhibitors"],
    "ERBB2": ["HER2-targeted therapy"],
    "EGFR": ["EGFR inhibitors"],
    "PDCD1": ["PD-1 immunotherapy"],
    "CD274": ["PD-L1 immunotherapy"]
}

drug_mapping_rows = []

for cluster, genes in cluster_gene_lists.items():
    for gene in genes:
        gene = str(gene).upper()
        if gene in drug_target_dict:
            for therapy in drug_target_dict[gene]:
                drug_mapping_rows.append({
                    "cluster": str(cluster),
                    "gene": gene,
                    "therapy": therapy
                })

drug_map_df = pd.DataFrame(drug_mapping_rows)
drug_map_df.to_csv("scvi_drug_target_mapping.csv", index=False)

print(drug_map_df.head(30))
print(drug_map_df.shape)
```

```
  cluster     gene                         therapy
0       0   COL1A1     Fibroblast/stroma targeting
1       0   COL1A2     Fibroblast/stroma targeting
2       2    KRT18         Epithelial tumor marker
3       2     KRT8         Epithelial tumor marker
4       2  TACSTD2        TROP2-targeted therapies
5       3    EPCAM  Epithelial-targeting therapies
6       6     CD3D            T-cell immunotherapy
7       8     CD68     Macrophage targeting (TAMs)
8      12   COL1A1     Fibroblast/stroma targeting
9      12   COL1A2     Fibroblast/stroma targeting
(10, 3)
```

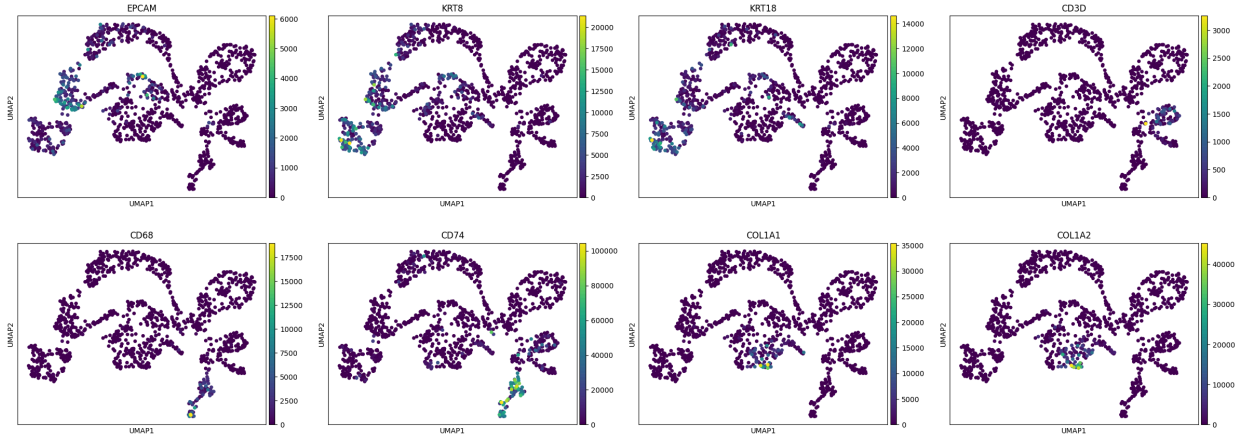
In [89]:
```
cluster_map_scvi = {
    "0": "Fibroblast/Stromal",
    "1": "Proliferating tumor",
    "2": "Tumor epithelial",
    "3": "T cells",
    "4": "Tumor subtype",
    "5": "Tumor subtype",
    "6": "Macrophages",
    "7": "Tumor subtype",
```

```python
    "8": "Antigen-presenting cells",
    "9": "Tumor subtype",
    "10": "Tumor subtype",
    "11": "Stress tumor",
    "12": "Tumor subtype"
}

summary_rows = []

for cluster in sorted(cluster_gene_lists.keys(), key=lambda x: int(x)):
    cluster_str = str(cluster)

    cell_type = cluster_map_scvi.get(cluster_str, "Unknown")
    cell_count = (adata_scvi.obs["leiden"].astype(str) == cluster_str).sum()
    top_genes = ", ".join([str(g).upper() for g in cluster_gene_lists[cluster][:5]])

    pathway_terms = top_pathways[
        top_pathways["cluster"].astype(str) == cluster_str
    ]["Term"].head(3).tolist()

    pathway_terms = ", ".join(pathway_terms) if pathway_terms else "NA"

    summary_rows.append({
        "cluster": cluster_str,
        "cell_type": cell_type,
        "cell_count": cell_count,
        "top_genes": top_genes,
        "top_pathways": pathway_terms
    })

cluster_summary_df = pd.DataFrame(summary_rows)
cluster_summary_df.to_csv("scvi_cluster_biological_summary.csv", index=False)

print(cluster_summary_df)
```

```
   cluster                 cell_type  cell_count  \
0        0        Fibroblast/Stromal         171
1        1       Proliferating tumor         139
2        2          Tumor epithelial         136
3        3                   T cells         120
4        4             Tumor subtype         118
5        5             Tumor subtype         108
6        6               Macrophages         103
7        7             Tumor subtype          82
8        8  Antigen-presenting cells          66
9        9             Tumor subtype          60
10      10             Tumor subtype          38
11      11              Stress tumor          35
12      12             Tumor subtype          31

                                top_genes  \
0   SERPING1, SPARCL1, SPARC, TIMP3, CALD1
1           ITGB8, LTF, PSMB4, TLK1, TRPS1
2       KRT18, AZGP1, KRT8, XBP1, KIAA1324
3          FAM60A, MGST1, DSC2, TRIM2, CD9
4         PPIL6, SHOX, STX1B, FLG2, EMX2OS
5    RNASE1, CALML5, TMSB10, KRT14, MAGEA3
6    IL2RG, PTPRC, TSC22D3, HLA-E, ARHGDIB
7      MS4A2, UTY, JAKMIP2, SREK1IP1, PAG1
8     FCER1G, LYZ, CD68, HLA-DRA, HLA-DRB1
9     TRA2A, HNRNPL, CTNNB1, TFAP2C, PADI2
10       NLRC5, CXCR4, HLA-A, PTPRC, HLA-B
11  PGM5P2, PCDH11X, CRX, TBXA2R, UGDH-AS1
12      TIMP1, C1S, AXL, CRISPLD2, RARRES2

                                         top_pathways
0   Extracellular Matrix Organization R-HSA-147424...
1   Post-translational Protein Modification R-HSA-...
2   Cytoplasmic Translation (GO:0002181), Peptide ...
3   Bundle Of His cell-Purkinje Myocyte Adhesion I...
4   Sodium/Proton Exchangers R-HSA-425986, POU5F1 ...
5   Negative Regulation Of Cysteine-Type Endopepti...
6   Immune System R-HSA-168256, Immunoregulatory I...
7   Response To Increased Oxygen Levels (GO:003629...
8   Immune System R-HSA-168256, Antigen Processing...
9   Metabolism Of Proteins R-HSA-392499, Epidermal...
10  Immune System R-HSA-168256, Human T-cell leuke...
11  Regulation Of Vasoconstriction (GO:0019229), C...
12  Regulation Of Complement Cascade R-HSA-977606,...
```

In [90]:

```
sc.pl.umap(
    adata_scvi,
    color=["EPCAM", "KRT8", "KRT18", "CD3D", "CD68", "CD74", "COL1A1", "COL1A2"],
    ncols=4
)
```



In [91]:
```
final_interpretation = """
scVI-based single-cell RNA-seq analysis identified multiple biologically distinct TNBC
including tumor epithelial cells, proliferating tumor cells, macrophages, T cells, ant
and fibroblast/stromal populations.

Pathway enrichment analysis highlighted cell-cycle activity, immune-related signaling,
supporting the presence of tumor heterogeneity and active tumor–microenvironment inter

Drug-target mapping linked selected marker genes to therapeutic strategies involving i
cell-cycle inhibition, macrophage targeting, and stromal targeting.
"""
print(final_interpretation)
```

```
scVI-based single-cell RNA-seq analysis identified multiple biologically distinct TNB
C cell populations,
including tumor epithelial cells, proliferating tumor cells, macrophages, T cells, an
tigen-presenting cells,
and fibroblast/stromal populations.

Pathway enrichment analysis highlighted cell-cycle activity, immune-related signalin
g, and stromal/ECM-associated biology,
supporting the presence of tumor heterogeneity and active tumor–microenvironment inte
ractions.

Drug-target mapping linked selected marker genes to therapeutic strategies involving
immunotherapy,
cell-cycle inhibition, macrophage targeting, and stromal targeting.
```

In [ ]: